E_n-CHIRAL ALGEBRAS, DERIVED CENTRES, AND THE OPERADIC CIRCLE

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Abstract. E_n-chiral algebras are factorisation algebras on configuration spaces of algebraic curves, inherently geometric. We construct bar complexes at every operadic level ($n = 1, 2, 3, \infty$) on every curve geometry (formal disk, punctured disk, annulus, nodal curves, pair of pants), and prove that the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ carries E_2 structure from E_1 input (Deligne) and E_3 from E_2 input (higher Deligne). The Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ governs the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$, not the bar complex itself. The topologisation theorem establishes $\mathsf{SC}^{\mathrm{ch,top}}$ + conformal vector $\simeq E_3^{\mathrm{top}}$ for affine Kac–Moody at non-critical level (chain-level) and, via Drinfeld–Sokolov reduction, for Virasoro and all $\mathcal{W}$-algebras (cohomological, with model-level $(E_3)_\infty$). The E_n operadic circle $E_3 \to E_2 \to E_1 \to E_2 \to E_3$ is proved to close for simple $\mathfrak{g}$ via the E_3 identification theorem. Five notions of E_1-chiral algebra are distinguished. The chiral quantum group equivalence identifies three structures (R-matrix, A_∞, coproduct) on the Koszul locus.

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Date: April 26, 2026.

1. E_n-CHIRAL ALGEBRAS ON CONFIGURATION SPACES

1.1. **The problem.** A chiral algebra is a factorisation algebra on a complex curve. The point of departure is the collision of labelled points: when $z_1 \to z_2$ on a curve X, the OPE extracts algebraic data from the singularity along the diagonal, and the bar construction records the iterated collisions as a coalgebra. The bar differential uses the logarithmic 1-form $\eta_{ij} = d\log(z_i - z_j)$ as propagator, the Arnold relation $\eta_{ij} \wedge \eta_{jk} + \eta_{jk} \wedge \eta_{ki} + \eta_{ki} \wedge \eta_{ij} = 0$ ensures $d^2 = 0$, and the whole construction lives on Fulton–MacPherson compactifications $\overline{C}_k(X)$ of ordered configurations.

This is the $n = 2$ entry of a larger pattern. Replace the curve by a real n-manifold, the logarithmic 1-form by a closed $(n-1)$-form, and the Arnold relation by its higher-dimensional analogue (the Totaro relation): the result is E_n-Koszul duality. The chiral bar complex of a vertex algebra is the holomorphic refinement of this topological construction, and the E_n hierarchy parametrises both the manifold dimension and the available algebraic structure at each level.

Two axes must be kept separate throughout this paper. The *chirality axis* is the hierarchy

$$E_\infty\text{-chiral} \subset P_\infty\text{-chiral} \subset E_1\text{-chiral}$$

of commutativity constraints on the OPE, all on a fixed curve. The Heisenberg, affine Kac–Moody, and Virasoro algebras are E_∞-chiral (local, symmetric OPE); the Yangian and Etingof–Kazhdan quantum vertex algebras are E_1-chiral (nonlocal, ordered OPE). The *dimensional axis* is the E_n ladder

$$n = 1 \text{ (interval)}, \quad n = 2 \text{ (surface/curve)}, \quad n = 3 \text{ (Chern–Simons)}, \quad \ldots$$

indexed by real manifold dimension. The chiral bar complex on a complex curve belongs to $n = 2$; the $n = 1$ case is the classical associative/A_∞ bar construction on intervals. An E_1-chiral algebra on a curve is *not* the same as an E_1-algebra in the topological sense: the former carries holomorphic structure on a 2-dimensional base, the latter lives on a 1-dimensional interval.

1.2. **Configuration spaces and their cohomology.**

Definition 1.1 (Configuration space)**.** For a smooth manifold M of real dimension n, the *ordered configuration space of k points* is

$$\mathrm{Conf}_k(M) = \{(x_1, \ldots, x_k) \in M^k : x_i \neq x_j \text{ for } i \neq j\}.$$

The *unordered configuration space* is the quotient $\mathrm{Conf}_k(M)/\Sigma_k$, where Σ_k acts by permuting the labels.

Theorem 1.2 (Arnold presentation; $n = 2$)**.** *The cohomology ring $H^*(\mathrm{Conf}_k(\mathbb{C}); \mathbb{Q})$ is the graded-commutative algebra generated by classes $\eta_{ij} \in H^1$, $1 \le i < j \le k$, subject to the* Arnold relations*:*

$$\eta_{ij} \wedge \eta_{jk} + \eta_{jk} \wedge \eta_{ki} + \eta_{ki} \wedge \eta_{ij} = 0 \qquad (i < j < k). \tag{1}$$

This is the Orlik–Solomon algebra OS_k [1].

Theorem 1.3 (Totaro presentation; general n [7, 26])**.** *For $n \ge 2$, the ring $H^*(\mathrm{Conf}_k(\mathbb{R}^n); \mathbb{Q})$ is generated by classes $\alpha_{ij} \in H^{n-1}$, $1 \le i \neq j \le k$, subject to:*

(i) (Symmetry) $\alpha_{ij} = (-1)^n \alpha_{ji}$.
(ii) (Totaro relations) $\alpha_{ij} \cdot \alpha_{jk} + \alpha_{jk} \cdot \alpha_{ki} + \alpha_{ki} \cdot \alpha_{ij} = 0$ *for distinct i, j, k.*
(iii) (Degree) $\alpha_{ij}^2 = 0$ *when n is odd; when n is even, α_{ij}^2 is determined by the Euler class.*

At $n = 2$, identifying $\mathbb{R}^2 \cong \mathbb{C}$, the Totaro relations specialize to the Arnold relations (1)*. At $n = 1$, the configuration space* $\mathrm{Conf}_k(\mathbb{R})$ *has contractible components indexed by orderings, and H^0 is the group algebra of Σ_k; the degree-$(n-1) = 0$ generators encode the ordered structure of the associative bar complex.*

Remark 1.4 (Arnold at $n = 2$ as Totaro specialization). The generators η_{ij} of Arnold are the Totaro generators α_{ij} at $n = 2$. The following table organises the connection between the E_n level, the manifold, and the bar complex:

n	Propagator degree	Manifold	Bar complex
1	0	interval/circle	associative bar (classical)
2	1	surface/curve	chiral bar (this paper)
3	2	3-manifold	Chern–Simons bar
n	$n-1$	n-manifold	E_n bar

1.3. **Fulton–MacPherson compactification.**

Definition 1.5 (FM compactification of $\mathrm{Conf}_k(\mathbb{R}^n)$). The *Fulton–MacPherson compactification* $\overline{\mathrm{Conf}}_k(\mathbb{R}^n)$ is the real oriented blowup of $(\mathbb{R}^n)^k$ along all partial diagonals, ordered by reverse inclusion. It is the closure of the image of

$$\mathrm{Conf}_k(\mathbb{R}^n) \hookrightarrow (\mathbb{R}^n)^k \times \prod_{i<j} S^{n-1} \times \prod_{|S|\geq 2} [0, \infty],$$

recording positions, relative directions $\theta_{ij} = (x_i - x_j)/|x_i - x_j|$, and relative distances. The result is a smooth manifold with corners.

Proposition 1.6 (FM operad [8]). *The codimension-1 boundary strata of $\overline{\mathrm{Conf}}_k(\mathbb{R}^n)$ are indexed by subsets $S \subset \{1, \ldots, k\}$ with $|S| \geq 2$, and satisfy*

$$\partial_S \overline{\mathrm{Conf}}_k(\mathbb{R}^n) \cong \overline{\mathrm{Conf}}_{k-|S|+1}(\mathbb{R}^n) \times \overline{\mathrm{Conf}}_{|S|}(\mathbb{R}^n).$$

This factorisation equips $\{\overline{\mathrm{Conf}}_k(\mathbb{R}^n)\}_{k\geq 0}$ with the structure of the FM operad FM_n, a model for E_n [12, 24].

1.4. **The E_n propagator.**

Definition 1.7 (E_n propagator). An *E_n propagator* on an oriented n-manifold M is a smooth closed $(n-1)$-form $G \in \Omega^{n-1}(M \times M \setminus \Delta)$ satisfying:

(i) (Poincaré dual of diagonal) $\int_{M\times M} G \wedge \pi_2^* \eta = \int_M \eta$ for every closed η.
(ii) (Singularity) Near Δ, in geodesic coordinates $r = x_1 - x_2$,

$$G \sim \frac{1}{\mathrm{vol}(S^{n-1})} \frac{\sum_{a=1}^{n} (-1)^{a-1} r_a \, dr_1 \wedge \cdots \wedge \widehat{dr_a} \wedge \cdots \wedge dr_n}{|r|^n} + \text{smooth}.$$

(iii) (Symmetry) $\sigma^* G = (-1)^n G$ under the swap $(x_1, x_2) \mapsto (x_2, x_1)$.

Example 1.8 ($n = 2$: the chiral propagator). For $M = \mathbb{C}$, the propagator is $G = \eta_{12} = d\log(z_1 - z_2)$, a meromorphic 1-form. Residues along $z_1 = z_2$ extract OPE data. The symmetry $\sigma^* G = d\log(z_2 - z_1) = G$ holds since $d\log(-1) = 0$.

Example 1.9 ($n = 3$: the Chern–Simons propagator). For $M = \mathbb{R}^3$, the propagator is the closed 2-form

$$G = \frac{1}{4\pi} \frac{\epsilon_{abc}\, r^a \, dr^b \wedge dr^c}{|r|^3}, \qquad r = x_1 - x_2.$$

Integration over S^2 gives the linking number: $\int_{S^2} G = 1$.

Example 1.10 ($n = 1$: the associative case). For $M = \mathbb{R}$, the "propagator" is a 0-form: the sign function $G(x_1, x_2) = \mathrm{sgn}(x_1 - x_2)$, which records the ordering of points on the real line. The bar complex is the classical tensor coalgebra $T^c(s^{-1}\bar{A})$ with deconcatenation coproduct; there is no de Rham differential to speak of, and the Arnold/Totaro relations are trivial (degree-0 generators commute vacuously).

Proposition 1.11 (Residue as linking sphere integral). *For an E_n propagator G and a smooth function f near the diagonal, the E_n residue is*

$$\operatorname{Res}_\Delta(G \wedge f) \;:=\; \lim_{\varepsilon\to 0}\int_{|x_1-x_2|=\varepsilon} G\cdot f(x_1,x_2) \;=\; f(x,x).$$

This generalises the chiral residue $\operatorname{Res}_{z_1=z_2}$ *at $n=2$.*

Proof. The singularity condition forces $G|_{|r|=\varepsilon}$ to be the normalised volume form on S^{n-1} up to terms vanishing as $\varepsilon \to 0$. The integral therefore extracts $f(x,x)$. □

1.5. The E_n bar complex.

Definition 1.12 (E_n bar complex). Let A be an E_n-algebra. The E_n *bar complex* is the chain complex

$$\bar{B}_{E_n}(A) \;=\; \bigoplus_{k\geq 1} \Omega^*\big(\overline{\mathrm{Conf}}_k(\mathbb{R}^n)\big) \otimes_{\Sigma_k} (s^{-1}\bar{A})^{\otimes k},$$

with differential

$$d_{E_n} \;=\; d_{\mathrm{dR}} \;+\; \sum_{\substack{S\subset\{1,\dots,k\}\\ |S|\geq 2}} \operatorname{Res}_{\partial_S}(G_S \wedge \mu_S),$$

where G_S is the product of propagators for pairs within S, $\operatorname{Res}_{\partial_S}$ is the linking-sphere residue along the boundary stratum ∂_S of $\overline{\mathrm{Conf}}_k(\mathbb{R}^n)$, and $\mu_S\colon \bar{A}^{\otimes|S|} \to \bar{A}$ is the E_n-multiplication.

Theorem 1.13 ($d^2=0$ from Totaro relations). *The differential d_{E_n} satisfies $d_{E_n}^2 = 0$.*

Proof. Decompose $d_{E_n}^2 = d_{\mathrm{dR}}^2 + [d_{\mathrm{dR}}, d_{\mathrm{res}}] + d_{\mathrm{res}}^2$. The first term vanishes. The cross-term vanishes because G is closed. The square d_{res}^2 involves iterated residues: for three points i,j,k, the three iterated residues cancel by the Totaro relation $\alpha_{ij}\cdot\alpha_{jk} + \alpha_{jk}\cdot\alpha_{ki} + \alpha_{ki}\cdot\alpha_{ij} = 0$. Nested collisions cancel by the E_n-algebra associativity coherences encoded in the FM operad structure (Proposition 1.6). □

Remark 1.14 (n-dependence of the bar complex). At $n=1$: the propagator has degree 0, the de Rham differential vanishes, and the bar complex reduces to $T^c(s^{-1}\bar{A})$ with the classical bar differential from the algebra multiplication. At $n=2$: the propagator is a 1-form, and the bar complex is the chiral bar complex $\bar{B}^{\mathrm{ch}}(\mathcal{A})$ of this paper. At $n=3$: the propagator is a 2-form, and the bar complex computes perturbative Chern–Simons invariants.

1.6. The four levels.

The operadic levels relevant to chiral algebras on curves are:

Definition 1.15 (Four operadic levels on a curve). Let X be a smooth complex curve. We distinguish four levels of algebraic structure:

(i) *E_∞-chiral* (local, symmetric OPE): the factorisation algebra is defined on unordered configuration spaces, with Σ_k-equivariant multiplication. All standard vertex algebras (Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}_N$, free-field algebras) are E_∞-chiral.

(ii) *E_2-topological*: the bar complex $\bar{B}_{E_2}(A)$ uses the topological propagator $(1/2\pi)\, d\arg(z_1 - z_2)$ and depends only on the oriented topology of X.

(iii) *E_1-chiral* (ordered, nonlocal OPE): the factorisation algebra lives on ordered configuration spaces $\mathrm{Conf}_k^{\mathrm{ord}}(X)$, with no Σ_k quotient. Yangians and Etingof–Kazhdan quantum vertex algebras are E_1-chiral.

(iv) E_3*-topological*: the algebra of bulk observables. Accessed via the derived centre (topologisation theorem, Section 5).

Remark 1.16 (The chirality axis is orthogonal to the dimensional axis). An E_∞-chiral algebra and an E_1-chiral algebra are both "$n = 2$ objects" in the dimensional ladder: they are both factorisation algebras on a complex curve. The distinction is the commutativity of the OPE, not the manifold dimension. The E_2-topological bar complex uses $n = 2$ topological configurations; the E_1 ordered bar complex uses the E_1-chiral ordering. These are different structures on the same underlying space.

1.7. **The Heisenberg at each level.** The Heisenberg algebra H_k at level k, generated by a single field $J(z)$ with OPE $J(z)J(w) \sim k/(z-w)^2$, provides the universal test case. As an abelian chiral algebra, its structure simplifies at every operadic level.

Example 1.17 (Heisenberg: E_∞-chiral)**.** The Heisenberg H_k is E_∞-chiral. The OPE has a double pole $k/(z-w)^2$ and no simple pole (the algebra is abelian), so all OPE products are Σ_k-symmetric. The E_∞-chiral bar complex is the coshuffle coalgebra $S^c(s^{-1}\overline{H}_k)$ with the bar differential encoding the double-pole OPE. The classical r-matrix is

$$r^{\mathrm{Heis}}(z) = k/z,$$

with $\kappa(H_k) = \mathrm{av}(r(z)) = k$. At $k = 0$, $r(z) = 0$ and $\kappa = 0$: the algebra becomes trivially abelian with no scattering.

Example 1.18 (Heisenberg: E_1-chiral)**.** The ordered bar complex $\bar{B}^{\mathrm{ord}}(H_k) = T^c(s^{-1}\overline{H}_k)$ has deconcatenation coproduct. Since H_k is E_∞, the formality bridge holds: $\bar{B}^{\mathrm{ord}}(H_k) \xrightarrow{\sim} \bar{B}^{\Sigma}(H_k)$ is a quasi-isomorphism. The R-matrix is trivial (the Heisenberg is commutative), and the averaging map $\mathrm{av}\colon \mathfrak{g}^{E_1}_{H_k} \to \mathfrak{g}^{\mathrm{mod}}_{H_k}$ loses nothing. The Heisenberg is class G (shadow tower terminates at degree 2: $\Delta = 8\kappa\, S_4 = 0$ because $S_4 = 0$).

Example 1.19 (Heisenberg: E_2-topological)**.** The E_2 bar complex $\bar{B}_{E_2}(H^{\mathrm{top}}_k)$ uses the topological propagator $(1/2\pi)\, d\arg(z_1 - z_2)$. The bar cohomology is concentrated in degree 1: the Heisenberg is Koszul. The Koszul dual is $H^!_k = \mathrm{Sym}^{\mathrm{ch}}(V^*)$, the symmetric chiral algebra on the dual generator. The Koszul complementarity is $\kappa(H_k) + \kappa(H^!_k) = k + (-k) = 0$.

Example 1.20 (Heisenberg: E_3-topological)**.** The derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k)$ carries E_2 from the chiral direction and E_1 from the topological direction. Since H_k has a conformal vector at any $k \neq 0$ (the Sugawara element $T = (1/2k) :JJ:$ at level k), topologisation applies: the BRST cohomology carries E^{top}_3. For the Heisenberg, this E_3 structure is trivial (all operations are determined by the commutative product and the linear Lie bracket), reflecting class G.

Remark 1.21 (The Heisenberg as Casimir–Goldstone opening). The Heisenberg is the abelian Kac–Moody algebra: the same object as the level-k U(1) current algebra, as the boundary algebra of abelian Chern–Simons theory, and as the universal target of the Casimir element in the chiral bar complex. Every formula in this paper is computed first for H_k and then extended to general families by variation of the r-matrix.

1.8. **Relation to prior work.** The topological foundation is factorisation homology of manifolds with E_n coefficients, developed by Ayala–Francis [2] and extended to structured manifolds in Ayala–Francis [3]. In the topological setting, the bar complex of an E_n-algebra A computes the factorisation homology $\int_M A$ when M is an n-framed manifold; the chiral bar complex of the present paper is the holomorphic refinement of this construction. Higher enveloping algebras, introduced by Knudsen [18], provide a Lie-algebraic model for factorisation homology of free E_n-algebras: the higher enveloping algebra $U_n(\mathfrak{g})$ of a Lie algebra computes the factorisation homology of the free E_n-algebra on $\mathfrak{g}$. The operadic bar-cobar machinery underlying the Koszul duality between E_n-algebras and E_n-coalgebras is developed in full generality in Fresse [10], where the homotopy theory of operads is constructed using Grothendieck–Teichmüller methods. The intrinsic formality of E_n-operads, proved by Fresse–Willwacher [11], is the technical input for arrow (4) of the operadic circle (5): it implies that the higher Deligne conjecture is controlled by the rational homotopy type of E_n-operads, without auxiliary choices. Higher Hochschild homology, developed by Ginot–Tradler–Zeinalian [14], computes the factorisation homology of E_∞-algebras over arbitrary simplicial sets; the chiral Hochschild theory of Section 3 is the algebro-geometric analogue, with simplicial sets replaced by algebraic curves.

2. Bar chain models on curve geometries

The E_n bar complex of Section 1 is defined on configuration spaces of $\mathbb{R}^n$: a flat, contractible base with no periods, no curvature, no modular anomaly. A chiral algebra lives on curves whose global topology shapes the bar differential. On the formal disk D, the bar complex is flat ($d^2 = 0$). On a genus-g curve Σ_g, the period matrix introduces curvature $d^2_{\mathrm{fib}} = \kappa \cdot \omega_g$. The passage from D to Σ_g is the passage from algebra to geometry.

2.1. The formal disk D.

Definition 2.1 (Bar complex on the formal disk)**.** Let $D = \mathrm{Spec}\,\mathbb{C}[[z]]$ be the formal disk and $\mathcal{A}$ a chiral algebra on D. The *bar complex on D* is

$$\bar{B}_D(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}), \qquad d_{\bar{B}} = \sum_{i<j} \mathrm{Res}_{z_i=z_j}\big(\eta_{ij} \wedge \mu_{ij}\big), \tag{2}$$

where $\eta_{ij} = d\log(z_i - z_j)$ is the chiral propagator, μ_{ij} is the OPE product at positions z_i, z_j, and $\bar{\mathcal{A}} = \ker(\varepsilon)$ is the augmentation ideal. The coproduct is deconcatenation:

$$\Delta[a_1|\cdots|a_n] \;=\; \sum_{i=0}^{n} [a_1|\cdots|a_i] \otimes [a_{i+1}|\cdots|a_n].$$

This makes $\bar{B}_D(\mathcal{A})$ an E_1-chiral coassociative coalgebra (a dg coalgebra over $(\mathrm{Ass}^{\mathrm{ch}})^!$). The bar complex on the formal disk sees genus 0 only. There is no period matrix, no curvature, no modular anomaly. The bar differential uses only the collision data of points on a contractible space.

Proposition 2.2 ($d^2 = 0$ on D)**.** *The differential* (2) *satisfies* $d^2_{\bar{B}} = 0$.

Proof. By Theorem 1.13 at $n = 2$: the Arnold relation on $\mathrm{Conf}_k(D)$ ensures cancellation of iterated residues. On the formal disk, the global topology is trivial and there is no curvature correction. □

Example 2.3 (Heisenberg on D)**.** The bar complex $\bar{B}_D(H_k)$ has generators $s^{-1}J_{-n}\,\mathbf{1}$ for $n \geq 1$, where $J_{-n} = (1/(n-1)!)\,\partial^{n-1}J$. The bar differential encodes the OPE $J(z)J(w) \sim k/(z-w)^2$: the only nonzero component of $d_{\bar{B}}$ at degree 2 is

$$d_{\bar{B}}[J_{-1}|J_{-1}] \;=\; k\,[J_{-2}],$$

the residue of the double-pole OPE contracted against $d\log(z_1 - z_2)$, which absorbs one pole to produce a simple-pole residue. Bar cohomology is concentrated in degree 1: the Heisenberg is Koszul.

2.2. The punctured disk D^*.

Definition 2.4 (Bar complex on D^*)**.** Let $D^* = \mathrm{Spec}\,\mathbb{C}((z))$ be the punctured formal disk. The bar complex on D^* is

$$\bar{B}_{D^*}(\mathcal{A}) \;=\; T^c(s^{-1}\bar{\mathcal{A}}), \qquad d_{\bar{B}} = d_D + \sum_i \oint_{|z_i|=\varepsilon} \eta_i \wedge \partial_i,$$

where d_D is the disk bar differential (2) and the second term records the monodromy around the puncture. The punctured disk is the home of the state-field correspondence: a bar element of degree k is a k-fold collision pattern that wraps around the origin.

Proposition 2.5 (The E_2-E_1 gap on D^*). *On the punctured disk, the ordered bar complex $\bar{B}^{\mathrm{ord}}_{D^*}(\mathcal{A})$ carries strictly more information than the symmetric bar $\bar{B}^{\Sigma}_{D^*}(\mathcal{A})$. The averaging map* $\mathrm{av}\colon \bar{B}^{\mathrm{ord}} \to \bar{B}^{\Sigma}$ *has kernel controlled by braid-group representations of $\pi_1(\mathrm{Conf}_k(D^*)) = B_k^{\mathrm{aff}}$ (the affine braid group).*

Proof. The fundamental group $\pi_1(\mathrm{Conf}_k(D^*))$ is the affine braid group on k strands, which surjects onto Σ_k with kernel the pure affine braid group. The symmetric bar complex takes Σ_k-coinvariants, killing the braid-group representations. For E_∞-chiral algebras (Heisenberg, Kac–Moody), the R-matrix is trivial and the loss is inessential. For E_1-chiral algebras (Yangian), the R-matrix is nontrivial and the kernel of av contains the KZ associator and its higher-degree descendants. □

Example 2.6 (Heisenberg on D^*). The monodromy of H_k around the puncture is trivial (the R-matrix $r(z) = k/z$ has no branch cut), so $\bar{B}_{D^*}(H_k) \simeq \bar{B}_D(H_k)$. The E_2-E_1 gap is zero: the ordered and symmetric bars agree.

2.3. The annulus.

Definition 2.7 (Bar complex on the annulus). Let $A_{r,R} = \{z : r < |z| < R\}$ be an annulus. The bar complex on $A_{r,R}$ computes the topological Hochschild homology of $\mathcal{A}$ as a chiral algebra:

$$\bar{B}_{A_{r,R}}(\mathcal{A}) \;\simeq\; \bar{B}(\mathcal{A}) \otimes^{\mathbf{L}}_{\mathcal{A}^e} \bar{B}(\mathcal{A}),$$

where $\mathcal{A}^e = \mathcal{A} \otimes \mathcal{A}^{\mathrm{op}}$ is the enveloping algebra and the derived tensor product computes the self-Tor of the bar resolution. The annulus has two boundary circles (inner and outer), giving it the structure of a bimodule over $\mathcal{A}$.

Proposition 2.8 (Annulus computes chiral Hochschild homology). *The factorisation homology of $\mathcal{A}$ over the annulus $A_{r,R}$ is*

$$\int_{A_{r,R}} \mathcal{A} \;\simeq\; \mathrm{ChirHoch}_*(\mathcal{A}).$$

The inner boundary gives the left $\mathcal{A}$-module structure, the outer boundary the right $\mathcal{A}$-module structure, and the factorisation homology of the annulus is the derived trace.

Proof. The annulus is the trace geometry for factorisation homology: $\int_{S^1\times I} \mathcal{A} \simeq \mathrm{HH}_*(\mathcal{A})$ by the Künneth property of factorisation homology applied to the product $S^1 \times I$, where S^1 contributes the trace and I the bimodule structure. The chiral refinement replaces the topological S^1 with the holomorphic annulus and the topological HH with ChirHoch. □

Example 2.9 (Heisenberg on the annulus). The chiral Hochschild homology of H_k is $\mathrm{ChirHoch}_*(H_k) \cong \mathbb{C}[J_{-1}, J_{-2}, \ldots]$, the polynomial algebra on the negative modes. The annulus partition function is the character $\mathrm{tr}_{H_k}(q^{L_0}) = 1/\eta(q)$, where $\eta(q) = q^{1/24} \prod_{n\geq 1}(1 - q^n)$ is the Dedekind eta function.

2.4. Nodal curves and curvature.

Definition 2.10 (Bar complex at genus g). Let Σ_g be a smooth genus-g curve and $\mathcal{A}$ a chiral algebra on Σ_g. The genus-g bar complex is

$$\bar{B}^{(g)}(\mathcal{A}) \;=\; \bigoplus_{k\geq 1} \Omega^*\big(\overline{\mathrm{C}}_k(\Sigma_g)\big) \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes k},$$

with differential $d_g = d_{\mathrm{dR}} + d_{\mathrm{res}}$, where d_{res} uses the genus-g propagator $d\log E(z,w)$, with $E(z,w)$ the prime form on Σ_g. The propagator $d\log E(z,w)$ has weight 1 for all g (the prime form is a section of $K^{-1/2} \boxtimes K^{-1/2}$).

Proposition 2.11 (Curvature at genus $g \geq 1$). *At genus $g \geq 1$, the bar differential is curved:*

$$d_{\mathrm{fib}}^2 \;=\; \kappa(\mathcal{A}) \cdot \omega_g,$$

where $\omega_g = c_1(\lambda) \in H^2(\overline{\mathcal{M}}_g)$ is the first Chern class of the Hodge line bundle on the moduli space $\overline{\mathcal{M}}_g$, and $\kappa(\mathcal{A})$ is the scalar curvature of $\mathcal{A}$. The curvature is on the fiberwise bar differential over $\overline{\mathcal{M}}_g$, not on the total bar differential (*which satisfies $d^2 = 0$ on the convolution algebra over $\overline{\mathcal{M}}_{g,n}$*).

Proof. The genus-g propagator $d\log E(z,w)$ acquires period corrections from $H^0(\Sigma_g, \Omega^1)$. These corrections introduce the curvature $\kappa \cdot \omega_g$ when the bar differential is restricted to fibers of the universal curve over $\overline{\mathcal{M}}_g$. The computation reduces to the anomaly of the Hodge line bundle and is performed in the convolution algebra formalism. □

Example 2.12 (Heisenberg at genus g). At genus $g \geq 1$, the curvature is $d_{\mathrm{fib}}^2 = k \cdot \omega_g$, using $\kappa(H_k) = k$. At $k = 0$, the curvature vanishes: the bar complex is uncurved at every genus. The genus-g anomaly $\mathrm{obs}_g = \kappa \cdot \lambda_g = k \cdot \lambda_g$ (uniform-weight) gives $F_1 = k/24$ from the Hodge class on $\overline{\mathcal{M}}_{1,1}$.

2.5. Pair of pants and fusion.

Definition 2.13 (Pair of pants). The pair of pants P is a genus-0 surface with three boundary circles. The factorisation homology over P computes the fusion product:

$$\int_P \mathcal{A} \;=\; \mathcal{A} \otimes^{\mathbf{L}}_{\mathcal{A}} \mathcal{A},$$

where the derived tensor product is taken over the chiral algebra itself, viewed as a bimodule via left and right OPE.

Proposition 2.14 (Fusion from bar complex). *The fusion product of two $\mathcal{A}$-modules M_1, M_2 is computed by the two-pointed bar complex:*

$$M_1 \otimes^{\mathbf{L}}_{\mathcal{A}} M_2 \;\simeq\; \bar{B}(\mathcal{A}; M_1, M_2) \;=\; \bigoplus_{k\geq 0} M_1 \otimes (s^{-1}\bar{\mathcal{A}})^{\otimes k} \otimes M_2,$$

with differential contracting adjacent factors via OPE. The associated derived category is the category of conformal blocks.

Example 2.15 (Heisenberg fusion). The Fock modules $\mathcal{F}_\lambda$ of H_k at momentum λ fuse by $\mathcal{F}_{\lambda_1} \otimes^{\mathbf{L}}_{H_k} \mathcal{F}_{\lambda_2} \simeq \mathcal{F}_{\lambda_1+\lambda_2}$, reflecting the additive structure of the abelian current algebra. The fusion rules are trivial: all Fock modules fuse freely. This is the class G triviality.

3. The derived chiral centre

3.1. The problem.

Given a chiral algebra $\mathcal{A}$, what is the algebra of bulk observables of the associated 3-dimensional holomorphic–topological theory? The answer is the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}) = C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$, the chiral Hochschild cochain complex. This is not the commutant of $\mathcal{A}$ in itself (the naive centre, which may be too small), but the derived endomorphism object computed via the bar resolution.

The key structural result is that this derived centre carries E_2 structure from the chiral direction (the chiral Deligne conjecture, Theorem 3.3 below) and E_1 structure from the topological direction, giving a total $\mathsf{SC}^{\mathrm{ch,top}}$ structure on the *pair* $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$.

3.2. Chiral Hochschild cochains.

Definition 3.1 (Chiral Hochschild cochain complex)**.** Let $\mathcal{A}$ be a chiral algebra on a curve X and let $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ denote the chiral endomorphism operad (the collection of multilinear maps $\mathcal{A}^{\otimes m} \to \mathcal{A}$ parametrised by points on X and their collision patterns). The *chiral Hochschild cochain complex* is

$$C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}) \;=\; \prod_{m \geq 0} \mathrm{Hom}_{\Sigma_m}\big(\overline{\mathrm{C}}_m(\mathbb{C}),\ \mathrm{Hom}(\mathcal{A}^{\otimes m}, \mathcal{A})\big), \tag{3}$$

with differential combining the bar differential of the configuration space with the internal differential of $\mathcal{A}$. A degree-k cochain is a map that takes m inputs from $\mathcal{A}$, arranged on the configuration space $\overline{\mathrm{C}}_m(\mathbb{C})$, and produces a single output in $\mathcal{A}$ with the correct equivariance.

Definition 3.2 (Chiral braces)**.** The cochain complex $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ carries *chiral brace operations*

$$\{f; g_1, \ldots, g_k\} \;\in\; C^{p+q_1+\cdots+q_k-k}_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$$

for $f \in C^p$ and $g_i \in C^{q_i}$, defined by inserting the g_i into the inputs of f at specified positions in $\overline{\mathrm{C}}_m(\mathbb{C})$. These are the chiral analogues of the classical Gerstenhaber braces. The brace operations satisfy:

(i) the pre-Lie identity on the single insertion $\{f; g\}$ (which defines the chiral Gerstenhaber bracket $[f, g] = \{f; g\} - (-1)^{(|f|-1)(|g|-1)}\{g; f\}$);
(ii) higher coherence relations making $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$ a homotopy E_2-algebra.

3.3. The chiral centre theorem.

Theorem 3.3 (Chiral centre theorem: the operadic centre of $\mathsf{SC}^{\mathrm{ch,top}}$ is $C^\bullet_{\mathrm{ch}}$)**.** *Let $\mathcal{A}$ be an E_∞-chiral algebra on a curve X and $A = \mathcal{A}|_{\{z_0\}}$ the fiber at a basepoint. The* operadic centre *of A in the Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ is quasi-isomorphic to the chiral Hochschild cochain complex as E_2-algebras:*

$$Z_{\mathsf{SC}^{\mathrm{ch,top}}}(A) \;\simeq\; C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}).$$

The E_2-structure comes from the closed sector $\{\overline{\mathrm{C}}_k(\mathbb{C})\}_{k\geq 0}$ of $\mathsf{SC}^{\mathrm{ch,top}}$: the FM compactification of the plane provides the operadic parameter space for the braces and the cup product.

Proof. By the operadic centre construction (Definition 3.4 below), an element of $Z_{\mathsf{SC}^{\mathrm{ch,top}}}(A)$ at degree (k, m) is a map

$$\phi_{k,m} \colon \overline{\mathrm{C}}_k(\mathbb{C}) \times E_1(m) \;\longrightarrow\; \mathrm{Hom}(A^{\otimes m}, A),$$

equivariant for Σ_k on $\overline{\mathrm{C}}_k(\mathbb{C})$ and Σ_m on $E_1(m)$. The $E_1(m)$ factor provides the open inputs; the $\overline{\mathrm{C}}_k(\mathbb{C})$ factor provides the closed/holomorphic inputs. Taking the limit over all m, the components assemble into the chiral Hochschild cochain complex (3). The closed-sector composition maps in $\mathsf{SC}^{\mathrm{ch,top}}$ (FM insertion in $\mathbb{C}$) produce the brace operations, and the compatibility conditions of the operadic centre are precisely the higher coherence relations that make the braces into a homotopy E_2-structure. □

Definition 3.4 (Operadic centre)**.** Let $\mathcal{O}$ be a two-coloured operad with colours $\{\mathsf{c}, \mathsf{o}\}$ and the vanishing constraint $\mathcal{O}(\ldots, \mathsf{o}, \ldots; \mathsf{c}) = \varnothing$ (no open-to-closed maps). For an $\mathcal{O}^{\mathsf{o}}$-algebra A, the *operadic centre* $Z_{\mathcal{O}}(A)$ is the terminal object in the category of $\mathcal{O}$-algebra pairs (B, A) with fixed open colour A:

$$Z_{\mathcal{O}}(A) \;=\; \varprojlim_{\substack{(B,A)\in \\ \mathcal{O}\text{-}\mathsf{Alg}_{/A}}} B.$$

Concretely, it is the equaliser in

$$Z_{\mathcal{O}}(A) \;\hookrightarrow\; \prod_{m\geq 0} \mathrm{Hom}_{\Sigma_m}\big(\mathcal{O}(\mathsf{c}, \mathsf{o}^m; \mathsf{o}),\ \mathrm{Hom}(A^{\otimes m}, A)\big) \;\rightrightarrows\; (\cdots),$$

where the two maps enforce closed-sector and open-sector equivariance.

3.4. Theorem H: concentration in degrees $\{0, 1, 2\}$.

Theorem 3.5 (Theorem H: cohomological concentration). *Let $\mathcal{A}$ be a Koszul chiral algebra. The chiral Hochschild cohomology* $\mathrm{ChirHoch}^*(\mathcal{A}) = H^*(C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A}))$ *is concentrated in cohomological degrees* $\{0, 1, 2\}$*:*

$$\mathrm{ChirHoch}^i(\mathcal{A}) = 0 \qquad \textit{for } i \geq 3.$$

The Hilbert series is polynomial. The three nonvanishing degrees have the following interpretations:

(i) $\mathrm{ChirHoch}^0(\mathcal{A}) = Z(\mathcal{A})$*: the naive centre.*
(ii) $\mathrm{ChirHoch}^1(\mathcal{A})$*: first-order deformations of $\mathcal{A}$ as a chiral algebra.*
(iii) $\mathrm{ChirHoch}^2(\mathcal{A})$*: obstructions to extending deformations.*

Remark 3.6 (Amplitude, not virtual dimension). Concentration in degrees $\{0, 1, 2\}$ is a statement about *cohomological amplitude*: the cochain complex is quasi-isomorphic to one supported in degrees 0, 1, 2. This is not the same as "virtual dimension 2" (an arithmetic invariant) or "$\mathrm{ChirHoch}^*(\mathcal{A}) = \mathbb{C}[\Theta]$" (which is false for all non-trivial algebras). It also differs from the Gelfand–Fuchs cohomology $H^*_{\mathrm{GF}}(\mathfrak{g})$, which is infinite-dimensional for the Virasoro algebra.

Example 3.7 (Heisenberg: $\mathrm{ChirHoch}^*(H_k)$). For the Heisenberg algebra H_k:

$$\begin{aligned} \mathrm{ChirHoch}^0(H_k) &= \mathbb{C} && \text{(the centre is 1-dimensional)},\\ \mathrm{ChirHoch}^1(H_k) &= \mathbb{C} && \text{(one-parameter deformation: the level } k\text{)},\\ \mathrm{ChirHoch}^2(H_k) &= \mathbb{C} && \text{(one obstruction class)}. \end{aligned}$$

Total dimension: $\dim \mathrm{ChirHoch}^*(H_k) = 3$. For comparison: the naive centre $Z(H_k) = \mathbb{C}$ has dimension 1; the derived centre has dimension 3, detecting the Ext^1 and Ext^2 contributions invisible to the commutant.

Example 3.8 (Affine Kac–Moody: $\mathrm{ChirHoch}^*(V_k(\mathfrak{g}))$). For the affine Kac–Moody algebra $V_k(\mathfrak{g})$ with $\mathfrak{g}$ simple of dimension $\dim(\mathfrak{g})$:

$$\begin{aligned} \mathrm{ChirHoch}^0(V_k(\mathfrak{g})) &= \mathbb{C},\\ \mathrm{ChirHoch}^1(V_k(\mathfrak{g})) &\cong \mathfrak{g} && (\dim(\mathfrak{g}) \text{ deformations}),\\ \mathrm{ChirHoch}^2(V_k(\mathfrak{g})) &= \mathbb{C}. \end{aligned}$$

Total dimension: $\dim(\mathfrak{g}) + 2$. The $\mathrm{ChirHoch}^1$ is isomorphic to $\mathfrak{g}$ as a vector space; the $\dim(\mathfrak{g})$ directions correspond to the current-algebra deformations $J^a(z) \mapsto J^a(z) + \epsilon\, \phi^a(z)$.

3.5. Three Hochschild theories: chiral, topological, categorical.

Warning 3.9 (Three Hochschild theories: chiral, topological, categorical). Three distinct Hochschild-type constructions appear in the E_n operadic circle: chiral, topological, and categorical. The geometry of the base space determines which of the chiral/topological/categorical theories applies.

(i) *Chiral Hochschild:* $\mathrm{ChirHoch}^*(\mathcal{A})$. Input: E_∞-chiral algebra on a curve X. Configuration complex: $\overline{\mathrm{C}}_k(X)$. Output: E_2 structure (chiral braces), concentrated in degrees $\{0, 1, 2\}$ (Theorem 3.5).
(ii) *Topological Hochschild:* $\mathrm{THH}(A)$. Input: E_1-algebra (associative). Configuration complex: $\mathrm{Conf}_k(\mathbb{R})$ (ordered, locally constant). Output: E_2 structure (classical Deligne conjecture).
(iii) *Categorical Drinfeld centre:* $Z(\mathcal{A}\text{-mod})$. Input: E_1-monoidal category of $\mathcal{A}$-modules. Output: E_2-braided monoidal category (the centre carries half-braidings).

Chiral Hochschild lives on algebraic curves; topological Hochschild lives on S^1; the categorical Drinfeld centre lives in the representation category. The three constructions give different answers for the same algebra, and their relationship is the content of the E_n operadic circle (Section 6).

3.6. The $\mathsf{SC}^{\mathrm{ch,top}}$ structure on the pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$.

Principle 3.10 (The bar complex is E_1-chiral coassociative; $\mathsf{SC}^{\mathrm{ch,top}}$ lives on the derived centre pair). The ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is an E_1-chiral coassociative coalgebra: a dg coalgebra over $(\mathrm{Ass}^{\mathrm{ch}})^!$ with differential from OPE residues and deconcatenation coproduct. It does *not* carry $\mathsf{SC}^{\mathrm{ch,top}}$ structure.

The $\mathsf{SC}^{\mathrm{ch,top}}$ structure emerges on the *derived chiral centre pair* $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$:

- The derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is the *closed colour* (bulk), carrying E_2 from $\overline{\mathrm{C}}_k(\mathbb{C})$.
- The chiral algebra $\mathcal{A}$ is the *open colour* (boundary), carrying E_1 from ordered configurations.
- The bulk acts on the boundary: the action maps $\alpha_{k,m} : \overline{\mathrm{C}}_k(\mathbb{C}) \times E_1(m) \times (\mathcal{Z}^{\mathrm{der}})^{\otimes k} \times \mathcal{A}^{\otimes m} \to \mathcal{A}$ are the mixed-sector operations.
- No boundary-to-bulk maps exist: $\mathsf{SC}^{\mathrm{ch,top}}(\ldots, \mathsf{o}, \ldots; \mathsf{c}) = \varnothing$.

The passage from bar complex to derived centre is the chiral Hochschild construction: the convolution $\mathrm{Hom}(\bar{B}^{\mathrm{ord}}(\mathcal{A}), \mathcal{A})$ is the computational model for $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$. The bar complex is the E_1 engine; the derived centre is the $\mathsf{SC}^{\mathrm{ch,top}}$ output.

Example 3.11 (Heisenberg: $\mathsf{SC}^{\mathrm{ch,top}}$ structure)**.** For H_k, the derived centre pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k), H_k)$ has closed colour $\mathcal{Z}^{\mathrm{der}} \simeq \mathbb{C}^3$ (Example 3.7) and open colour H_k. The bulk-to-boundary action is determined by the identity map (degree 0), the level deformation (degree 1), and the obstruction class (degree 2). All higher mixed-sector operations vanish: the Heisenberg is SC-formal.

4. The Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$

4.1. Definition.

Definition 4.1 (Holomorphic–topological Swiss-cheese operad)**.** The two-coloured topological operad $\mathsf{SC}^{\mathrm{ch,top}}$ has colours $\{\mathsf{ch}, \mathsf{top}\}$ (closed/holomorphic and open/topological) with three sectors:

(i) *Closed sector* (all inputs and output closed/holomorphic):
$$\mathsf{SC}^{\mathrm{ch,top}}((\mathsf{ch}, \ldots, \mathsf{ch}); \mathsf{ch}) \;=\; \overline{\mathrm{C}}_k(\mathbb{C}).$$
This is the E_2-operad parametrising holomorphic configurations on the complex plane.

(ii) *Open sector* (all inputs and output open/topological):
$$\mathsf{SC}^{\mathrm{ch,top}}((\mathsf{top}, \ldots, \mathsf{top}); \mathsf{top}) \;=\; E_1(m).$$
This is the E_1-operad of ordered configurations on the real line.

(iii) *Mixed sector* (k closed and m open inputs, open output):
$$\mathsf{SC}^{\mathrm{ch,top}}((\mathsf{ch}^k, \mathsf{top}^m); \mathsf{top}) \;=\; \overline{\mathrm{C}}_k(\mathbb{C}) \times E_1(m).$$
No mixed operations with closed output: the closed colour receives no information from the open colour.

Composition is componentwise: FM insertion in $\mathbb{C}$, interval insertion in E_1. The no-open-to-closed rule is the directionality constraint $\mathsf{SC}^{\mathrm{ch,top}}(\ldots, \mathsf{top}, \ldots; \mathsf{ch}) = \varnothing$.

Remark 4.2 ($\mathsf{SC}^{\mathrm{ch,top}}$ is not E_3). The Swiss-cheese operad is two-coloured with a strict directionality constraint. Dunn additivity ($E_n \otimes E_m \simeq E_{n+m}$) applies to *single-coloured* operads. The product structure $\overline{\mathrm{C}}_k(\mathbb{C}) \times E_1(m)$ in the mixed sector does *not* imply that $\mathsf{SC}^{\mathrm{ch,top}}$ is $E_2 \otimes E_1 = E_3$: the directionality constraint (no open-to-closed) breaks the symmetry required for Dunn's theorem. The passage from $\mathsf{SC}^{\mathrm{ch,top}}$ to E_3 requires additional data (Section 5).

4.2. Koszul duality of $\mathsf{SC}^{\mathrm{ch,top}}$.

Theorem 4.3 (Koszulity of $\mathsf{SC}^{\mathrm{ch,top}}$ [22]). *The operad $\mathsf{SC}^{\mathrm{ch,top}}$ is Koszul.*

Proposition 4.4 (Koszul dual cooperad: three sectors). *The Koszul dual cooperad $(\mathsf{SC}^{\mathrm{ch,top}})^{!}$ has three sectors:*

(i) Closed: $\dim(\mathsf{SC}^{\mathrm{ch,top}})^{!}(n;\mathsf{c}) = (n-1)!$ (Lie *cooperad, since* $\mathrm{Com}^{!} = \mathrm{Lie}$).
(ii) Open: $\dim(\mathsf{SC}^{\mathrm{ch,top}})^{!}(m;\mathsf{o}) = m!$ (Ass *cooperad, self-Koszul-dual*).
(iii) Mixed *(k closed, m open inputs):* $\dim(\mathsf{SC}^{\mathrm{ch,top}})^{!}(k,m;\mathsf{o}) = (k-1)!\binom{k+m}{m}$.

Proof. The closed sector is $\mathrm{Com}^{!} = \mathrm{Lie}$ with $\dim \mathrm{Lie}(n) = (n-1)!$. The open sector is $\mathrm{Ass}^{!} = \mathrm{Ass}$ with $\dim = m!$. The mixed sector combines a Lie factor $(k-1)!$ from the holomorphic direction with a binomial coefficient $\binom{k+m}{m}$ from the (k,m)-shuffle interleaving of k closed inputs among m ordered open inputs on $\overline{\mathrm{C}}_k(\mathbb{C}) \times E_1(m)$. □

Corollary 4.5 ($\mathsf{SC}^{\mathrm{ch,top}}$ is not Koszul self-dual). *The dimensions of $\mathsf{SC}^{\mathrm{ch,top}}$ and $(\mathsf{SC}^{\mathrm{ch,top}})^{!}$ disagree in the closed sector:*

$$\dim\ \mathsf{SC}^{\mathrm{ch,top}}(n;\mathsf{c}) = 1 \quad \textit{(Com)}, \qquad \dim(\mathsf{SC}^{\mathrm{ch,top}})^{!}(n;\mathsf{c}) = (n-1)! \quad \textit{(Lie)}.$$

Therefore $\mathsf{SC}^{\mathrm{ch,top}} \ncong (\mathsf{SC}^{\mathrm{ch,top}})^{!}$*: the operad is* not *Koszul self-dual.*

The duality functor *on* $\mathsf{SC}^{\mathrm{ch,top}}$*-algebras is an involution (*$(P^{!})^{!} \cong P$ *for Koszul P), but the operad itself is not fixed. Koszul duality exchanges* Com *(closed) with* Lie *(closed) while preserving* Ass *(open).*

4.3. The five presentations of $\mathsf{SC}^{\mathrm{ch,top}}$.

An $\mathsf{SC}^{\mathrm{ch,top}}$-algebra pair $(\mathcal{A}^{\mathrm{cl}}, A^{\mathrm{op}})$ can be presented in five equivalent ways:

Proposition 4.6 (Five presentations of $\mathsf{SC}^{\mathrm{ch,top}}$-algebra structure). *The following structures on a pair $(\mathcal{A}^{\mathrm{cl}}, A^{\mathrm{op}})$ are equivalent for Koszul $\mathsf{SC}^{\mathrm{ch,top}}$:*

(i) Operadic: *a map of two-coloured operads* $\mathsf{SC}^{\mathrm{ch,top}} \longrightarrow \mathrm{End}_{(\mathcal{A}^{\mathrm{cl}},A^{\mathrm{op}})}$.
(ii) Koszul dual: *an* $(\mathsf{SC}^{\mathrm{ch,top}})^{!}$*-coalgebra structure on the pair* $(\bar{B}(\mathcal{A}^{\mathrm{cl}}), \bar{B}(A^{\mathrm{op}}))$*. The closed colour is a Lie coalgebra* $(\bar{B}_{\mathrm{Lie}}(\mathcal{A}^{\mathrm{cl}}))$*; the open colour is an associative coalgebra* $(\bar{B}_{\mathrm{Ass}}(A^{\mathrm{op}}))$.
(iii) Factorisation: *a factorisation algebra on* $\mathbb{C}\times\mathbb{R}^{\geq 0}$ *(the upper half-space), with the interior (*$\mathbb{C}\times\mathbb{R}^{>0}$*) carrying the closed colour and the boundary (*$\mathbb{C}\times\{0\}$*) carrying the open colour.*
(iv) BV/BRST: *a BV algebra whose classical master equation encodes the* $\mathsf{SC}^{\mathrm{ch,top}}$ *structure, with the antibracket splitting into holomorphic (closed) and topological (open) components.*
(v) Convolution: *a Maurer–Cartan element in the* $\mathsf{SC}^{\mathrm{ch,top}}$*-convolution algebra* $\mathfrak{g}_{\mathcal{A}}^{\mathsf{SC}^{\mathrm{ch,top}}} = \mathfrak{g}_{\mathcal{A}}^{\mathrm{mod}} \times \mathfrak{g}_{\mathcal{A}}^{\mathbb{R}}$*, where* $\mathfrak{g}^{\mathrm{mod}}$ *controls the closed direction (shadows* κ, $\mathfrak{C}$, $\mathfrak{Q}$*) and* $\mathfrak{g}^{\mathbb{R}}$ *controls the open direction (r-matrix, KZ associator, Yangian deformation).*

Remark 4.7 (Pentagon of equivalences). The five presentations are pairwise equivalent, but all equivalences route through the operadic hub (presentation (i)). No single pentagon theorem closes the circle of all five equivalences directly. Three links have independent mathematical content: Koszul–BV (the BV formalism realises the Koszul dual cooperad), BV–Convolution (the antibracket encodes the convolution Lie bracket), and Factorisation–Convolution (the global sections functor converts a factorisation algebra to a convolution algebra).

4.4. The Koszul dual algebra $\mathcal{A}^!$ as an $(\mathsf{SC}^{\mathrm{ch,top}})^!$-algebra.

Proposition 4.8 (Koszul dual carries $(\mathsf{SC}^{\mathrm{ch,top}})^! = (\mathrm{Lie}, \mathrm{Ass})$ structure)**.** *Let $\mathcal{A}$ be a Koszul chiral algebra. The Koszul dual $\mathcal{A}^! = ((\mathcal{A}^i)^\vee)$ carries the structure of an $(\mathsf{SC}^{\mathrm{ch,top}})^!$-algebra:*

(i) Closed colour: *the Sklyanin bracket $\{-,-\}$ makes $\mathcal{A}^!$ a Lie algebra (from $\mathrm{Com}^! = \mathrm{Lie}$).*
(ii) Open colour: *the Yangian product makes $\mathcal{A}^!$ an associative algebra (from $\mathrm{Ass}^! = \mathrm{Ass}$).*

This is an $(\mathsf{SC}^{\mathrm{ch,top}})^!$-algebra $= (\mathrm{Lie}, \mathrm{Ass})$-algebra, not *an $\mathsf{SC}^{\mathrm{ch,top}}$-algebra $= (\mathrm{Com}, \mathrm{Ass})$-algebra.*

Example 4.9 (Heisenberg: $H_k^!$)**.** The Koszul dual $H_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ has $\kappa(H_k^!) = -k$, so the Koszul complementarity gives $\kappa(H_k) + \kappa(H_k^!) = k + (-k) = 0$. The Lie bracket (Sklyanin bracket) on $H_k^!$ is trivial (abelian), reflecting the class G status. The Ass structure is the symmetric algebra product, which is commutative and thus carries no ordered data beyond the symmetric quotient.

5. The topologisation theorem

5.1. The problem.

The chiral centre theorem (Theorem 3.3) produces E_2 structure on the derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$, with the E_2 coming from holomorphic configurations $\overline{\mathrm{C}}_k(\mathbb{C})$. This E_2 is *holomorphic*: it depends on the complex structure of the curve. The dimensional ladder places E_3 at the level of 3-manifolds (Chern–Simons). How does the holomorphic E_2 centre upgrade to topological E_3?

The answer requires extra structure: an inner conformal vector. The conformal vector makes holomorphic translations Q-exact in the BRST complex, killing the dependence on the complex structure at the cohomological level. The BRST cohomology of the factorisation algebra on $\mathbb{C}$ becomes locally constant; a locally constant factorisation algebra on $\mathbb{R}^2$ is an E_2^{top}-algebra. Combined with the E_1 from the topological direction, Dunn additivity gives $E_2^{\mathrm{top}} \otimes E_1^{\mathrm{top}} = E_3^{\mathrm{top}}$.

5.2. The conformal vector.

Definition 5.1 (Inner conformal vector)**.** Let $\mathcal{A}$ be a chiral algebra and Q a BRST differential on the observables of a holomorphic–topological theory on $\mathbb{C} \times \mathbb{R}$ whose boundary chiral algebra is $\mathcal{A}$. An *inner conformal vector* for $(\mathcal{A}, Q)$ is a conformal vector $T(z) \in \mathcal{A}$ that is Q-exact in the algebra of bulk observables: there exists $G(z)$ (the antighost contraction) such that

$$T(z) \;=\; [Q, G(z)]$$

in BRST cohomology. The consequence: holomorphic translations $\partial_z \mathcal{O} = [T, \mathcal{O}] = [Q, [G, \mathcal{O}]]$ are Q-exact for every bulk observable $\mathcal{O}$.

Example 5.2 (Affine Kac–Moody: the Sugawara element)**.** For $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, the Sugawara element

$$T_{\mathrm{Sug}} \;=\; \frac{1}{2(k+h^\vee)} \sum_a {:}J^a J_a{:}$$

provides the inner conformal vector. The central charge is $c = k\dim(\mathfrak{g})/(k+h^\vee)$, and the antighost contraction is

$$G(z) \;=\; \frac{1}{2(k+h^\vee)} \sum_a {:}J^a(z)\,\bar{c}_a(z){:}, \tag{4}$$

where $\bar{c}_a$ is the antighost field of the BRST complex.

Example 5.3 (Heisenberg: the Sugawara element)**.** For H_k at $k \neq 0$, the Sugawara element is $T = (1/2k)\,{:}JJ{:}$ with central charge $c = 1$. At $k = 0$, the Sugawara element is undefined (the denominator $2k$ vanishes) and the algebra has no conformal vector: topologisation fails. The critical level for the Heisenberg is $k = 0$ (there is no dual Coxeter number for $\mathrm{U}(1)$).

Remark 5.4 (Critical level: topologisation fails). At the critical level $k = -h^\vee$ for affine Kac–Moody, the Sugawara denominator $2(k + h^\vee)$ vanishes: the Sugawara element is undefined, ∂_z is not Q-exact, and the factorisation algebra retains genuine holomorphic dependence. The E_2 structure remains holomorphic, not topological, and the upgrade to E_3 fails. The critical level is the precise obstruction to topologisation.

5.3. The topologisation theorem.

Theorem 5.5 (Cohomological topologisation for affine Kac–Moody). *Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra and $V_k(\mathfrak{g})$ the universal affine vertex algebra at non-critical level $k \neq -h^\vee$. The Sugawara element $T_{\mathrm{Sug}} = (1/2(k + h^\vee)) \sum_a :J^a J_a:$ provides an inner conformal vector. The derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$, which carries E_2 from the closed sector of* $\mathsf{SC}^{\mathrm{ch,top}}$ *(Theorem 3.3), has the following topologisation package:*

(i) (Cohomological E_3^{top}.) *The BRST cohomology $H^\bullet_Q(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})))$ carries an E_3^{top}-algebra structure:*
$$H^\bullet_Q(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))) \ \text{ is an } \ E_2^{\mathrm{top}} \otimes E_1^{\mathrm{top}} = E_3^{\mathrm{top}}\text{-algebra.}$$

(ii) (Chain-level model.) *The cohomology complex $M_k = H^\bullet_Q(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}})$, with zero differential and the E_3^{top} operations of part* (i)*, is a chain-level E_3^{top}-algebra quasi-isomorphic to the derived centre. The homotopy transfer theorem controls the dependence on the chosen retract: transferred $(E_3)_\infty$-models are unique up to ∞-quasi-isomorphism.*

(iii) (Original-complex lift: conditional.) *If there exists a family $G = (G_1, G_2, \dots)$ with G_1 the Sugawara antighost contraction* (4) *and $[m, G] = \partial_z$ in the brace deformation complex, then the original complex $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ itself admits a chain-level E_3^{top}-structure.*

Proof. Part (i). The inner conformal vector identifies $T_{\mathrm{Sug}} = [Q, G]$ in BRST cohomology. For any bulk observable $\mathcal{O}$, $\partial_z \mathcal{O} = [T_{\mathrm{Sug}}, \mathcal{O}] = [Q, [G, \mathcal{O}]]$, so ∂_z is Q-exact. The BRST cohomology of the factorisation algebra on $\mathbb{C}$ is therefore locally constant: it does not detect the complex structure. By the recognition theorem of Lurie [15], a locally constant factorisation algebra on $\mathbb{R}^2$ is an E_2^{top}-algebra. The E_1^{top} from the $\mathbb{R}$-direction (the open colour of $\mathsf{SC}^{\mathrm{ch,top}}$) combines with the E_2^{top} via Dunn additivity to give E_3^{top}.

Part (ii). Over $\mathbb{C}$, choose a splitting of cycles and boundaries in $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}$; this gives a strong deformation retract onto the cohomology complex M_k. Part (i) equips M_k with E_3^{top}. The homotopy transfer theorem for operadic algebras ensures that transferred $(E_3)_\infty$-models are unique up to ∞-quasi-isomorphism.

Part (iii). The condition $[m, G] = \partial_z$ asserts that holomorphic translations are null-homotopic in the brace deformation complex by an A_∞-coherent family of higher homotopies. Under this condition, the locally-constant recognition applies to the original cochain complex (not just to cohomology), and Dunn additivity produces E_3^{top} at chain level.

Critical level. At $k = -h^\vee$, the Sugawara denominator vanishes: the inner conformal vector is undefined, ∂_z is not Q-exact, and the factorisation algebra retains holomorphic dependence. The E_2 remains holomorphic, and the upgrade to E_3 fails. □

Remark 5.6 (The conformal vector is necessary). Without a conformal vector, the $\mathsf{SC}^{\mathrm{ch,top}}$-algebra produces a centre with holomorphic E_2 that genuinely detects the complex structure. The $\mathsf{SC}^{\mathrm{ch,top}}$ structure is the *intermediary* between E_1-chiral and E_3; the conformal vector is the bridge. The E_3 is E_3^{top} (topological), not E_3-chiral: the conformal vector kills the chiral direction.

Remark 5.7 (Scope of topologisation). Theorem 5.5 is proved for affine Kac–Moody $V_k(\mathfrak{g})$ at non-critical level, where the Sugawara element provides an explicit inner conformal vector and the 3d holomorphic–topological theory is Chern–Simons.

For Virasoro and $\mathcal{W}$-algebras, the cohomological E_3^{top} is now proved via DS reduction (Theorem 5.10), and the model-level $(E_3)_\infty$ follows (Proposition 5.13). The original-complex chain-level lift remains open for class M:

Conjecture 5.8 (Original-complex topologisation for class M)**.** *Let $\mathcal{A}$ be a chiral algebra with conformal vector $T(z)$ at non-critical level, and suppose the associated holomorphic–topological theory on $\mathbb{C} \times \mathbb{R}$ has a BRST differential Q with $T = [Q, G]$ for some antighost contraction G. Then:*

(i) (Cohomological.) $H_Q^\bullet(\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}))$ *carries an E_3^{top}-algebra structure. This is now a theorem for all $\mathcal{W}$-algebras obtained by DS reduction* (*Theorem 5.10*) *and all conformal vertex algebras with freely generated PVA associated graded* (*Khan–Zeng* [17]).

(ii) (Original-complex.) *The E_3^{top} structure lifts from BRST cohomology to the original cochain complex $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$. This remains open for class M (Virasoro, $\mathcal{W}_N$): the A_∞-coherence tower is infinite and the gauge rectification does not terminate. The coderived category is expected to absorb the obstruction.*

5.4. **The Heisenberg: topologisation.**

Example 5.9 (Heisenberg topologisation)**.** For H_k at $k \neq 0$: the Sugawara element $T = (1/2k) :JJ:$ provides the inner conformal vector. The 3d holomorphic–topological theory is abelian Chern–Simons on $\mathbb{C} \times \mathbb{R}$. All three parts of Theorem 5.5 hold unconditionally:

(i) BRST cohomology is E_3^{top} (the abelian case is trivial: all operations are determined by the commutative product and the Lie bracket).

(ii) The chain-level model is the 3-dimensional cochain complex $\mathbb{C} \oplus \mathbb{C}[-1] \oplus \mathbb{C}[-2]$ with trivial differential and E_3 structure from the abelian Chern–Simons.

(iii) The original-complex lift is unconditional because the A_∞ coherences are trivial (class G: all higher operations vanish).

At $k = 0$: the Sugawara element is undefined, and the algebra is stuck at $\mathsf{SC}^{\mathrm{ch,top}}$. The derived centre $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(H_0)$ carries holomorphic E_2 but not E_3^{top}.

5.5. **Topologisation via Drinfeld–Sokolov reduction.** The cohomological topologisation of Theorem 5.5 was stated for affine Kac–Moody. For Virasoro and $\mathcal{W}$-algebras, the conjecture (Conjecture 5.8) asked for the 3d HT BRST complex. The Costello–Gaiotto theorem [6] provides exactly this: every $\mathcal{W}$-algebra $\mathcal{W}^k(\mathfrak{g}, f)$ arises as the boundary chiral algebra of holomorphic Chern–Simons with Drinfeld–Sokolov boundary conditions. The 3d *bulk* is the same holomorphic Chern–Simons for $\mathfrak{g}$, independent of the nilpotent f. The DS stress tensor decomposes as $T_{\mathrm{DS}} = T_{\mathrm{Sug}} + T_{\mathrm{imp}}$, where the improvement term involves only Cartan currents: $T_{\mathrm{imp}} = \sum_{i,j}(x_0)_i\,(\kappa^{-1})^{h_i h_j}\,\partial J_{h_j}$ with $x_0 = \frac{1}{2}h_0 \in \mathfrak{h}$ the neutral element of the $\mathfrak{sl}_2$-triple (e, h_0, f). Since each $J_a = [Q_{\mathrm{CS}}, \bar{c}_a]$ on Q_{CS}-cohomology (by the BRST transformation of the antighost), both summands are Q_{CS}-exact in the 3d bulk.

Theorem 5.10 (Cohomological topologisation via DS reduction)**.** *Let $\mathfrak{g}$ be a finite-dimensional simple Lie algebra, $f \in \mathfrak{g}$ any nilpotent element, and $k \neq -h^\vee$. The $\mathcal{W}$-algebra $\mathcal{W} = \mathcal{W}^k(\mathfrak{g}, f)$ obtained by Drinfeld–Sokolov reduction from $V_k(\mathfrak{g})$ admits an E_3^{top}-structure on the BRST cohomology of its derived chiral centre $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{W})$:*

$$H_Q^\bullet(\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{W})) \;\; \textit{is an} \;\; E_3^{\mathrm{top}}\textit{-algebra.}$$

In particular, this applies to: Virasoro $\mathrm{Vir}_c = \mathcal{W}^k(\mathfrak{sl}_2, f_{\mathrm{prin}})$, principal $\mathcal{W}_N = \mathcal{W}^k(\mathfrak{sl}_N, f_{\mathrm{prin}})$ for all N, and the Bershadsky–Polyakov algebra $\mathcal{W}^k(\mathfrak{sl}_3, f_{\mathrm{min}})$.

Proof. The Costello–Gaiotto theorem [6] provides the 3d holomorphic–topological theory: holomorphic Chern–Simons for $\mathfrak{g}$ at level k, with DS boundary conditions determined by f. The 3d bulk is independent of f.

The DS-transported antighost is $G' = G_{\mathrm{Sug}} + G_{\mathrm{imp}}$, where $G_{\mathrm{Sug}} = (1/2(k+h^\vee)) \sum_a {:}J^a\, \bar{c}_a{:}$ and $G_{\mathrm{imp}} = -\sum_{i,j} (x_0)_i\, (\kappa^{-1})^{h_i h_j}\, \partial \bar{c}_{h_j}$. Both are well-defined bulk operators.

The Sugawara term: $[Q_{\mathrm{CS}}, G_{\mathrm{Sug}}] = T_{\mathrm{Sug}}$ on Q_{CS}-cohomology (the same identity as in Theorem 5.5).

The improvement term: Since $x_0 = \frac{1}{2}h_0 \in \mathfrak{h}$ (the Jacobson–Morozov theorem places h_0 in the Cartan subalgebra for any nilpotent), the improvement involves only Cartan currents. The identity $J_a = [Q_{\mathrm{CS}}, \bar{c}_a]$ on Q_{CS}-cohomology implies $\partial J_{h_j} = [Q_{\mathrm{CS}}, \partial \bar{c}_{h_j}]$. Hence $T_{\mathrm{imp}} = [Q_{\mathrm{CS}}, G_{\mathrm{imp}}]$.

BRST identity: $[Q_{\mathrm{CS}}, G'] = T_{\mathrm{Sug}} + T_{\mathrm{imp}} = T_{\mathrm{DS}}$ on Q_{CS}-cohomology. By the argument of Theorem 5.5(i): holomorphic translations are Q-exact, the factorisation algebra on $\mathbb{C}$ is locally constant on Q-cohomology, and Dunn additivity produces $E_2^{\mathrm{top}} \otimes E_1^{\mathrm{top}} = E_3^{\mathrm{top}}$. □

Remark 5.11 (Specialisation to $\mathfrak{sl}_2$). For $\mathfrak{g} = \mathfrak{sl}_2$ with $h^\vee = 2$, $\mathrm{rk} = 1$, $\rho^\vee = 1/2$, and $\kappa^{hh} = 1/2$, the DS-transported antighost reduces to

$$G'(z) \;=\; \frac{1}{2(k+2)}\big({:}J^e\,\bar{c}_f{:} + {:}J^f\,\bar{c}_e{:} + \tfrac{1}{2}{:}J^h\,\bar{c}_h{:}\big) + \tfrac{1}{4}\,\partial\bar{c}_h(z),$$

and the BRST identity reads $[Q_{\mathrm{CS}}, G'] = T_{\mathrm{DS}} = T_{\mathrm{Sug}} + \frac{1}{2}\partial J^h$ on Q_{CS}-cohomology. The improvement antighost $\frac{1}{4}\partial\bar{c}_h$ is the source of the $\frac{1}{2}\partial J^h$ term in the Miura transformation.

Remark 5.12 (The proof is cohomological). The BRST identity is proved on Q_{CS}-cohomology: the ghost bilinear in $Q_{\mathrm{CS}} \cdot \bar{c}_a$ is Q-exact but not zero, so at the cochain level $[Q_{\mathrm{CS}}, G']$ differs from T_{DS} by Q-exact terms. For topologisation, the cohomological identity suffices. The chain-level question is addressed in Proposition 5.13 below.

5.6. Chain-level $(E_3)_\infty$ for Virasoro and $\mathcal{W}$-algebras.

The cohomological topologisation of Theorem 5.10 does not address whether the E_3^{top}-structure lifts from BRST cohomology to the cochain complex itself. For affine Kac–Moody (class L), such a lift is proved in Volume I: the gauge rectification procedure terminates in finitely many steps because the shadow depth $r_{\max} = 3$ is finite. For Virasoro (class M), the shadow depth is infinite, and the gauge rectification does *not* terminate. The chain-level E_3 on the *original* brace deformation complex of $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$ is a genuinely harder problem.

There are, however, two unconditional chain-level results that can be extracted from the existing machinery.

Proposition 5.13 (Chain-level $(E_3)_\infty$ for Virasoro and $\mathcal{W}$-algebras). *Let $\mathcal{W} = \mathcal{W}^k(\mathfrak{g}, f)$ be a $\mathcal{W}$-algebra obtained by DS reduction from $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, including* $\mathrm{Vir}_c = \mathcal{W}^k(\mathfrak{sl}_2, f_{\mathrm{prin}})$. *The derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{W})$ admits a chain-level $(E_3)^{\mathrm{top}}_\infty$-algebra structure on a quasi-isomorphic model, in two independent ways.*

(i) (From cohomological topologisation.) *Theorem* 5.10 *produces the cohomological E_3^{top} on $M = H^\bullet_Q(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{W}))$. Over $\mathbb{C}$, choose a splitting of cycles and boundaries in the cochain complex to obtain a strong deformation retract from $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{W})$ onto M. The homotopy transfer theorem for E_3-algebras produces a transferred $(E_3)^{\mathrm{top}}_\infty$ structure on M* (*zero differential*)*, unique up to ∞-quasi-isomorphism.*

(ii) (From the 3d bulk.) *The 3d holomorphic Chern–Simons bulk for $\mathfrak{g}$ at level k has local observables that carry a chain-level E_3^{top}-structure, independent of the boundary condition: for affine Kac–Moody boundary, this is the content of the gauge rectification in Volume I. The DS boundary condition determines a BRST-type differential Q_{DS} on the bulk cochain complex C that is a derivation of the E_3^{top}*

structure on C, because Q_{DS} is a component of the full BV-BRST differential Q_{CS} and the E_3 structure was constructed to be compatible with Q_{CS}. The BRST cohomology $H^\bullet(C, Q_{\mathrm{DS}}) \cong \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{W})$ inherits $(E_3)^{\mathrm{top}}_\infty$ by homotopy transfer through the strong deformation retract from (C, Q_{DS}) to its cohomology.

This second route is independent of the proof of Theorem 5.10: it derives the E_3 structure on the $\mathcal{W}$-algebra derived centre from the already proved *chain-level E_3 on the Kac–Moody-side bulk, without passing through the DS-transported antighost.*

Proof. Part (i) is a direct application of the homotopy transfer theorem for operadic algebras [23]: the E_3^{top} operations on the cohomology complex M are strict, the strong deformation retract transfers them to $(E_3)^{\mathrm{top}}_\infty$ on the zero-differential model, and the transferred structure is unique up to ∞-quasi-isomorphism by the standard contractibility of the space of choices.

Part (ii) requires two inputs.

Input 1: Chain-level E_3 on the bulk. The gauge rectification for $V_k(\mathfrak{g})$ at non-critical level produces a corrected antighost contraction $\widetilde{G}_1$ satisfying $[Q_{\mathrm{CS}}, \widetilde{G}_1] = T_{\mathrm{Sug}}$ on the original BRST complex, and the brace defect $D^{(1)} = [m, \widetilde{G}_1] - \partial_z$ vanishes because the shadow depth $r_{\max} = 3$ is finite (class L). This gives chain-level E_3^{top} on the local bulk observables C as a factorisation algebra on $\mathbb{C} \times \mathbb{R}$.

Input 2: Q_{DS} is an E_3-derivation. The DS boundary condition is implemented at the 3d level by restricting to a sub-BV complex determined by the nilpotent f. The differential Q_{DS} is a component of the full BV-BRST differential Q_{CS} of the 3d bulk. The chain-level E_3 structure on C was constructed to be compatible with Q_{CS} (the brace equation $[m, \widetilde{G}_1] = \partial_z$ holds in the complex of Q_{CS}-cochains). Since Q_{DS} is a summand of Q_{CS}, compatibility descends: Q_{DS} acts as an E_3-derivation on C.

With Inputs 1 and 2, the homotopy transfer theorem for operadic algebras produces a transferred $(E_3)^{\mathrm{top}}_\infty$-structure on the Q_{DS}-cohomology $H^\bullet(C, Q_{\mathrm{DS}}) \cong \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{W})$. The transferred structure is unique up to ∞-quasi-isomorphism. □

Remark 5.14 (Three layers of chain-level control). Proposition 5.13 gives chain-level $(E_3)_\infty$ on the zero-differential cohomology complex. This is the model-level statement: sufficient for applications that depend only on the ∞-quasi-isomorphism class (existence of an E_3 enhancement, factorisation homology up to equivalence, deformation functors).

A stronger statement would be chain-level E_3 on the *original* cochain complex $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathrm{Vir}_c)$ itself. This requires an A_∞-coherent null-homotopy of ∂_z in the brace deformation complex, which for class M involves an infinite coherence tower (all $m_k^{\mathsf{SC}^{\mathrm{ch,top}}}$ are nontrivial). The tower does not terminate, and the coderived category is the expected framework for absorbing it.

The three layers are therefore:

(a) *Cohomological E_3^{top}*: proved unconditionally (Theorem 5.10).
(b) *Model-level $(E_3)^{\mathrm{top}}_\infty$*: proved unconditionally (Proposition 5.13).
(c) *Original-complex E_3^{top}*: open, obstructed by the infinite class M coherence tower. The coderived category is expected to absorb this obstruction.

For affine Kac–Moody (class L), all three layers are proved, because the coherence tower terminates at depth 3.

Remark 5.15 (The two obstructions to original-complex E_3 for Virasoro). For Vir_c specifically, the obstruction to layer (c) has two components:

(i) *The infinite shadow tower.* Virasoro is class M: all A_∞ operations m_k for $k \geq 2$ are nontrivial, producing an infinite coherence tower in the brace deformation complex. The gauge rectification procedure generates corrections at every degree, none of which vanishes on the ordinary cochain complex.

(ii) *The class M BV/bar discrepancy.* The chain-level BV = bar comparison fails for class M on ordinary chain complexes (the quartic discrepancy δ_4 is not a coboundary). In the coderived category, δ_4 is absorbed.

These are distinct obstructions: (i) is about the A_∞ coherence of the antighost homotopy, and (ii) is about the comparison between BV and bar constructions. Both point to the coderived category as the resolution.

5.7. **The iterated-Sugawara ladder and E_∞-topologisation.** Theorem 5.5 promotes $\mathsf{SC}^{\mathrm{ch,top}}$ to E_3^{top} by a single inner conformal vector $T(z)$ with antighost $G(z)$ satisfying $T = [Q, G]$. When the chiral algebra carries a *tower* of inner higher-spin Casimir fields $\{T^{(n)}\}_{2 \le n \le N+1}$, each admitting its own BRST primitive, the topologisation iterates: a depth-k inner stress tower promotes $\mathsf{SC}^{\mathrm{ch,top}}$ to E_{k+2}^{top}. The principal $\mathcal{W}_N$-algebra $\mathcal{W}_N$ realises exactly a depth-$(N-1)$ tower, and the limit $\mathcal{W}_\infty$ realises E_∞-topologisation.

Theorem 5.16 (Iterated Sugawara construction). *Let $\mathcal{A}$ be a chiral algebra carrying a higher-spin Casimir tower $\{T^{(n)}\}_{2 \le n \le N+1}$ of inner fields (so each $T^{(n)} \in \mathcal{A}$ and the OPE closes among them). Suppose each $T^{(n)}$ admits a BRST primitive $G^{(n)} \in \mathcal{A}$ with $T^{(n)} = [Q_{\mathrm{tot}}, G^{(n)}]$ on the BRST cohomology of the 3d holomorphic–topological theory associated with $\mathcal{A}$. Then the depth-k sub-tower $\{T^{(2)}, \dots, T^{(k+1)}\}$ topologises $\mathsf{SC}^{\mathrm{ch,top}}$ to E_{k+2}^{top}:*

$$H_Q^\bullet\big(\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})\big) \quad \textit{is an} \quad E_{k+2}^{\mathrm{top}}\textit{-algebra.}$$

Proof sketch. For each n the pair $(T^{(n)}, G^{(n)})$ makes the corresponding C-translation $\partial_{C,n}$ equal to $[Q, G^{(n)}]$, hence Q-exact on cohomology. A single Q-exact translation collapses one holomorphic direction into a topological one (Theorem 5.5); k independent such collapses collapse k holomorphic directions. Dunn additivity for the orthogonal E_1^{top} factors then assembles $E_2^{\mathrm{chiral}} \otimes \big(E_1^{\mathrm{top}}\big)^{\otimes k} \simeq E_{k+2}^{\mathrm{top}}$, giving the claimed structure. □

Corollary 5.17 (Specialisations). (i) *Virasoro* Vir_c *($N = 2$, depth 1): E_3^{top}.* (ii) *Principal $\mathcal{W}_N$ (depth $N-1$): E_{N+1}^{top}.* (iii) *Principal $\mathcal{W}_\infty[\Upsilon]$ (depth ∞): E_∞^{top}.*

Remark 5.18 (The operadic spiral). The iterated-Sugawara ladder and the E_n operadic circle (5) together realise an infinite bidirectional spiral. The bar construction descends the operadic level by one; the derived centre ascends it by one. The iterated Sugawara collapses any number of holomorphic directions into topological ones. Meeting at E_∞, the spiral records that 3d quantum gravity in Vol II is the $N = 2$ shadow of a $3d + \infty$ topological theory.

5.8. **Chiral higher Deligne and the $\mathsf{SC}^{\mathrm{ch,top}}$ heptagon.** The operadic circle closes through the *chiral higher Deligne theorem*: the derived chiral centre $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A})$ is the universal E_3-chiral algebra on which the Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ acts.

Theorem 5.19 (Chiral higher Deligne; Vol II). *Let $\mathcal{A}$ be an E_1-chiral algebra on a smooth curve X. The derived chiral centre $\mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(\mathcal{A}) = \mathrm{ChirHoch}^\bullet(\mathcal{A}, \mathcal{A})$ is the universal E_3-chiral algebra acted on by $\mathsf{SC}^{\mathrm{ch,top}}$ through the heptagon edges (3) $\leftrightarrow$ (4) $\leftrightarrow$ (5) of the five-presentation pentagon of $\mathsf{SC}^{\mathrm{ch,top}}$-algebras. The concentration theorem for $\mathrm{ChirHoch}^\bullet(\mathcal{A})$ in degrees $\{0, 1, 2\}$ is a consequence of E_3-rigidity at a point together with PBW collapse.*

Remark 5.20 (The seven faces of $\mathsf{SC}^{\mathrm{ch,top}}$). The pentagon of five $\mathsf{SC}^{\mathrm{ch,top}}$-algebra presentations (operadic, Koszul, factorisation, BV–BRST, convolution) extends to a heptagon when two further faces are included: (6) the Drinfeld-centre face $Z(\mathrm{Rep}_{\mathrm{fact}}(\mathcal{A})) \simeq \mathrm{Rep}_{\mathrm{fact}}(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}))^{E_2}$ through categorified bar–cobar with half-braiding, and (7) the derived-algebraic-geometry face through PTVV on $\mathrm{Map}(X \times \mathbb{R}_{\geq 0}, \mathrm{B}\,\mathrm{SC}\text{-Alg})$. The seven-edge closed heptagon records that $\mathsf{SC}^{\mathrm{ch,top}}$ is the generic $\mathsf{SC}^{\mathrm{ch,top}}$-algebra presentation; topologisation to E_3^{top} is proved for affine Kac–Moody at non-critical level, and is conjectural in general pending the chain-level lift of Conjecture 5.8.

5.9. **Classification by topologisation status.**

Proposition 5.21 (Topologisation status of standard families).

Family	*Conformal vector*	*Topologisation*	*Status*
H_k, $k \neq 0$	$T = (1/2k) :JJ:$	E_3^{top}	*proved*
$V_k(\mathfrak{g})$, $k \neq -h^\vee$	T_{Sug}	E_3^{top}	*proved (chain-level)*
$V_{-h^\vee}(\mathfrak{g})$ *(critical)*	*none*	$\mathsf{SC}^{\mathrm{ch,top}}$	*stuck*
Vir_c	$T(z)$	E_3^{top}	*proved (cohom. + model)*
$\mathcal{W}_N$	$T(z)$	E_3^{top}	*proved (cohom. + model)*
$\mathcal{W}^k(\mathfrak{g}, f)$ *(any f)*	T_{DS}	E_3^{top}	*proved (cohom. + model)*
H_0	*none*	$\mathsf{SC}^{\mathrm{ch,top}}$	*stuck*
Lattice VA, $k \neq 0$	$T = (1/2k) :JJ:$	E_3^{top}	*proved (abelian CS)*

The column "Status" records the strongest chain-level control: proved (chain-level) *means the E_3^{top} structure lifts to the original BRST complex (affine Kac–Moody, class L);* proved (cohom. + model) *means cohomological E_3^{top} (Theorem 5.10) and $(E_3)_\infty^{\mathrm{top}}$ on a quasi-isomorphic model (Proposition 5.13), but the original-complex lift remains open (class M: infinite coherence tower).*

6. The E_n operadic circle

6.1. **The circle.** The five theorems of modular Koszul duality live in the symmetric bar $\bar{B}^{\Sigma}(\mathcal{A})$: they are invariants surviving the averaging map $\mathrm{av}\colon \bar{B}^{\mathrm{ord}}(\mathcal{A}) \to \bar{B}^{\Sigma}(\mathcal{A})$. The E_n operadic circle explains why the averaging map exists, what it loses, and how the lost information can in principle be recovered. The circle is a sequence of five functorial operations, each shifting the operadic level by one:

$$E_3^{\mathrm{top}}(\text{bulk}) \xrightarrow{(1)\ \text{restrict}} E_2(\text{boundary chiral}) \xrightarrow{(2)\ \bar{B}^{\mathrm{ord}}} E_1(\text{bar/QG}) \xrightarrow{(3)\ Z(-)} E_2(\text{Drinfeld centre}) \xrightarrow{(4)\ \mathrm{HH}^*} E_3^{\mathrm{top}}(\text{derive} \tag{5}$$

Definition 6.1 (Five arrows of the operadic circle). The five arrows are:

(1) *Restriction to a codimension-2 defect.* A bulk E_3^{top}-algebra of observables on a 3-manifold M^3 restricts to a factorisation algebra on a codimension-2 defect curve $C \subset M^3$. The normal bundle $N_{C/M}$ of rank 2 supplies two directions: one holomorphic (along the curve) and one topological (transverse). The restricted algebra is E_2-chiral on C.

(2) *Ordered bar complex.* The E_2-chiral algebra on C passes to its ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$: an E_1-chiral coassociative coalgebra with deconcatenation coproduct, R-matrix, and (on the Koszul locus) Yangian structure.

(3) *Drinfeld centre.* The functor $Z\colon E_1\text{-}\mathsf{Cat} \to E_2\text{-}\mathsf{Cat}$ recovers an E_2-braided monoidal category from E_1-module data. This is the categorified analogue of the averaging map. The Drinfeld centre is the *sole source of nontrivial braiding*: direct restriction from E_3^{top} to E_2 gives only symmetric braiding, because $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial.

(4) *Higher Deligne conjecture.* The topological Hochschild cohomology of an E_2-algebra carries E_3^{top}-structure. This is the higher Deligne conjecture, proved by Francis [9]: $\mathrm{HH}^*(A)$ for an E_n-algebra A carries an E_{n+1}-structure.

(5) *Closing.* The fifth arrow closes the circle: the E_3^{top}-algebra obtained by traversing arrows (1)–(4) should be equivalent to the original bulk E_3^{top}-algebra. This is the content of the closing conjecture (Section 11).

Theorem 6.2 (Arrow 3 as the sole source of braiding)**.** *Let A be an E_3^{top}-algebra.*

(i) *Restriction of A to a codimension-2 locus gives an E_2-algebra $A|_C$ whose braiding is* symmetric*: the monodromy representation factors through $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3)) = \{1\}$, the trivial group.*

(ii) *The Drinfeld centre $Z(\mathcal{A}\text{-mod})$ of the E_1-monoidal category of $\mathcal{A}$-modules carries a half-braiding that is generically* non-symmetric*: the braiding comes from the R-matrix of the ordered bar complex, which records the nontrivial monodromy of $\pi_1(\mathrm{Conf}_2(\mathbb{C})) = \mathbb{Z}$ (the braid group on 2 strands).*

(iii) *The non-symmetric braiding in the Drinfeld centre is therefore not inherited from the original E_3 restriction; it is created by arrow 3. For E_∞-chiral algebras, the braiding is symmetric* (*the R-matrix is trivial*)*. For E_1-chiral algebras, the braiding is non-symmetric and encodes the KZ associator $\Phi_{\mathrm{KZ}} \in \exp(\hat{\mathfrak{t}}_3)$.*

Proof. Part (i): the fundamental group $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial because $\mathbb{R}^3 \setminus \{0\}$ is simply connected (the linking sphere S^2 carries no 1-cycles). The representation of the braid group on the E_2-restricted modules therefore factors through Σ_2: the braiding is a symmetric involution.

Part (ii): the Drinfeld centre $Z(\mathcal{A}\text{-mod})$ consists of pairs (M, γ) where $\gamma_N : M \otimes N \xrightarrow{\sim} N \otimes M$ is a half-braiding natural in N. When $\mathcal{A}$ is the module category of the ordered bar complex, the half-braiding is conjugation by the R-matrix: $\gamma_N = R_{M,N} \circ \tau$. For E_1-chiral algebras with nontrivial R-matrix, the braiding is non-symmetric.

Part (iii): since E_3 restriction gives only symmetric braiding and the Drinfeld centre produces non-symmetric braiding, the non-symmetric part is created by arrow 3. This is the mathematical content of the statement that the Drinfeld centre is the sole source of nontrivial braiding. □

6.2. **The $\mathsf{SC}^{\mathrm{ch,top}}$ intermediary.** Between the E_1-chiral world and the E_3^{top} bulk sits the Swiss-cheese operad $\mathsf{SC}^{\mathrm{ch,top}}$ (Section 4). The passage from $\mathsf{SC}^{\mathrm{ch,top}}$ to E_3^{top} requires an inner conformal vector (Section 5). Without such a vector, the $\mathsf{SC}^{\mathrm{ch,top}}$-algebra cannot topologise: the two colours (holomorphic and topological) remain distinct, and Dunn additivity does not apply.

Proposition 6.3 (The $\mathsf{SC}^{\mathrm{ch,top}}$ intermediary in the circle)**.** *The E_n operadic circle factors through $\mathsf{SC}^{\mathrm{ch,top}}$:*

$$E_3^{\mathrm{top}} \xrightarrow{(1)} E_2^{\mathrm{ch}} \xrightarrow{(2)} E_1^{\mathrm{ch}} \xrightarrow{\mathsf{SC}^{\mathrm{ch,top}}} (E_2^{\mathrm{ch}},\, E_1^{\mathrm{top}}) \xrightarrow{\nu} E_3^{\mathrm{top}}.$$

The penultimate stage is the $\mathsf{SC}^{\mathrm{ch,top}}$-pair $(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}), \mathcal{A})$. The last arrow requires the inner conformal vector ν; when ν exists, the two SC colours collapse to a single E_3^{top} structure.

Proof. The Drinfeld centre (arrow 3) and the higher Deligne conjecture (arrow 4) combine to produce the $\mathsf{SC}^{\mathrm{ch,top}}$-pair: the derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ carries the closed E_2 from the FM configuration spaces, and $\mathcal{A}$ carries the open E_1 from ordered configurations. The topologisation theorem (Theorem 5.5) provides the final arrow. □

6.3. **The formality bridge and its failure.**

Theorem 6.4 (Formality bridge for E_∞-chiral algebras). *For an E_∞-chiral algebra $\mathcal{A}$ (the entire standard families: Heisenberg, affine Kac–Moody, Virasoro, $\mathcal{W}_N$, and all free-field algebras), the averaging map is a quasi-isomorphism:*

$$\mathrm{av}\colon \bar{B}^{\mathrm{ord}}(\mathcal{A}) \xrightarrow{\sim} \bar{B}^{\Sigma}(\mathcal{A}).$$

The R-matrix is trivial (the braiding on the ordered bar is symmetric), the Yangian is cocommutative, and the averaging map loses nothing.

Proof. The E_∞-chiral condition means the OPE is Σ_k-equivariant: the factorisation $\mathcal{D}$-module descends to the symmetric Ran space. The R-matrix $S(z) = \mathrm{id}$ is the identity, and the Σ_k-action on the ordered bar complex is free. The averaging map takes Σ_k-coinvariants of a free action, which is a quasi-isomorphism by the standard acyclicity of the group algebra. □

Theorem 6.5 (Formality failure for E_1-chiral algebras). *For a genuinely E_1-chiral algebra (Yangian, Etingof–Kazhdan quantum vertex algebra), the averaging map is* not *a quasi-isomorphism:*

$$\dim H^n(\bar{B}^{\mathrm{ord}}(\mathcal{A})) = n! \quad \text{at degree } n, \qquad \dim H^n(\bar{B}^{\Sigma}(\mathcal{A})) = 1.$$

The ordered bar carries Arnold braid-group representations (from $\pi_1(\mathrm{Conf}_k(\mathbb{C})) = B_k$) that the symmetric bar annihilates.

Proof. The E_1-chiral condition means the R-matrix $S(z) \neq \mathrm{id}$ is a nontrivial solution of the quantum Yang–Baxter equation. The Σ_k-action on the ordered bar complex is twisted by $S(z)$, and the kernel of av records the braid-group representations. The dimension $n!$ at degree n is the dimension of the regular representation of Σ_n; the coinvariant quotient collapses to dimension 1. □

Remark 6.6 (What the five theorems see and what they miss). The five theorems of modular Koszul duality (A, B, C, D, H) extract the Σ_n-invariant content of the ordered bar: the scalar κ, given by $\mathrm{av}(r(z))$ in abelian and scalar families and by $\mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$ for non-abelian affine Kac–Moody, the genus anomaly $\mathrm{obs}_g = \kappa \cdot \lambda_g$ (uniform-weight), and the shadow tower. For E_∞-chiral algebras this is the complete picture. For E_1-chiral algebras, the five theorems see only the shadow, while the ordered bar carries the R-matrix, the KZ associator, and the full braid-group representation.

7. The E_3 identification theorem

7.1. **The problem.** The topologisation theorem (Theorem 5.5) produces an E_3^{top}-algebra on the BRST cohomology of the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ for simple $\mathfrak{g}$ at non-critical level. This is a formal deformation of the Chevalley–Eilenberg cochains $C^*(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$, parametrised by $\lambda = k + h^\vee$. There is a second source of E_3^{top}-algebras: the perturbative quantisation of Chern–Simons theory on $\mathbb{R}^3$ with gauge algebra $\mathfrak{g}$, carried out by Costello–Francis–Gwilliam [5]. Both are deformations of the same classical limit. Do they coincide?

7.2. **The P_3 reduction.**

Theorem 7.1 (E_3 formality [11, 20, 25]). *The operad E_3 is formal over $\mathbb{Q}$: there is a zig-zag of quasi-isomorphisms of operads*

$$E_3 \xleftarrow{\sim} \cdots \xrightarrow{\sim} H_*(E_3) = \mathsf{P}_3.$$

An E_3-algebra structure on a cochain complex is therefore determined, up to homotopy, by the induced P_3-algebra structure on cohomology.

Remark 7.2 (From E_3 to P_3). The P_3-algebra structure on $H^*(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))) \cong C^*(\mathfrak{g})$ consists of:

(i) a graded-commutative product (the cup product, $\mathrm{Sym}(\mathfrak{g}^\vee[-1])$);

(ii) a degree-$(1-3) = -2$ Lie bracket $\{-,-\}$ (the Schouten–Nijenhuis bracket on polyvector fields, or equivalently the linearisation of the Chern–Simons interaction).

The P_3 bracket is determined by its restriction to generators: $\{\phi_a, \phi_b\} = \lambda f_{ab}{}^c \phi_c$, where $\lambda = k + h^\vee$ and $f_{ab}{}^c$ are the structure constants of $\mathfrak{g}$. The coefficient λ comes from the formal disk computation: the chiral derived centre at level k deforms the classical cochains by exactly $\lambda = k + h^\vee$ times the Cartan 3-form.

7.3. **The identification.**

Theorem 7.3 (Identification of E_3-deformation families for simple $\mathfrak{g}$). *Let $\mathfrak{g}$ be a simple finite-dimensional Lie algebra, $V_k(\mathfrak{g})$ the universal affine vertex algebra at non-critical level $k \neq -h^\vee$, and $\lambda = k + h^\vee$. Then the derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))$ and the Costello–Francis–Gwilliam E_3-algebra $\mathcal{A}^\lambda$ from perturbative Chern–Simons are isomorphic as formal deformation families of E_3-algebras:*

$$\left\{\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))\right\}_{k+h^\vee \in \lambda\mathbb{C}[[\lambda]]} \simeq \left\{\mathcal{A}^\lambda\right\}_{\lambda \in \lambda H^3(\mathfrak{g})[[\lambda]]}. \tag{6}$$

Proof. The proof proceeds in four steps.

Step 1: E_3 formality reduces to P_3. By Theorem 7.1, both sides are determined by their P_3-structures on the common cohomology $C^*(\mathfrak{g}) = \mathrm{Sym}(\mathfrak{g}^\vee[-1])$. A formal deformation of a P_3-algebra with cohomology $\mathrm{Sym}(V[-1])$ is classified, order by order in λ, by the cohomology group $H^*(\mathrm{Sym}(V[-1]), \mathrm{Sym}(V[-1]))$ in the P_3-deformation complex. For the Lie algebra cochains, this deformation complex is controlled by $H^3(\mathfrak{g})$ at leading order.

Step 2: one-dimensionality of $H^3(\mathfrak{g})$. For $\mathfrak{g}$ simple, $H^3(\mathfrak{g};\mathbb{C})$ is one-dimensional, generated by the Cartan 3-form $\omega_3(X,Y,Z) = \langle X, [Y,Z]\rangle$, where $\langle\cdot,\cdot\rangle$ is the Killing form. The deformation base is therefore one-dimensional at each order in λ: any P_3-deformation of $C^*(\mathfrak{g})$ is determined by a single scalar at each order.

Step 3: formal disk comparison. Both deformations restrict to the formal disk $D = \mathrm{Spec}\,\mathbb{C}[[z]]$. On D, the chiral derived centre is computed from the bar complex $\bar{B}_D(V_k(\mathfrak{g}))$ (Definition 2.1), and the CFG algebra is computed from the 1-loop Feynman integral on $\mathbb{R}^3$ localised near a point. Both produce the same P_3 bracket $\{\phi_a, \phi_b\} = \lambda f_{ab}{}^c \phi_c$ on generators. Since the deformation at each order is controlled by a single scalar (Step 2), and the formal disk determines that scalar (by evaluation at a point), the two families coincide order by order.

Step 4: global extension. The local isomorphism (Step 3) extends to a global isomorphism of filtered E_3-algebras because both deformations are defined over the same one-dimensional base $\lambda H^3(\mathfrak{g})[[\lambda]]$ and agree at each order. The filtration is by powers of λ; the isomorphism is the identity at each graded piece. $\square$

Remark 7.4 (The role of $\dim H^3(\mathfrak{g}) = 1$). The proof uses $\dim H^3(\mathfrak{g}) = 1$ at Step 2 in an essential way. For non-simple $\mathfrak{g}$ (e.g. $\mathfrak{g} = \mathfrak{gl}_N = \mathfrak{sl}_N \oplus \mathfrak{gl}_1$), the deformation space is multi-dimensional: $\dim \mathrm{Sym}^2(\mathfrak{g}^*)^{\mathfrak{g}} = r + \binom{\dim \mathfrak{z}+1}{2}$, where r is the number of simple summands and $\mathfrak{z}$ is the centre. The formal disk comparison must then determine all independent scalars, which requires matching all independent invariant bilinear forms (e.g. $B_{\mathrm{tr}}(X,Y) = \mathrm{tr}(XY)$ and $B_{\mathrm{ab}}(X,Y) = \mathrm{tr}(X)\mathrm{tr}(Y)$ for $\mathfrak{gl}_N$). The identification extends to $\mathfrak{gl}_N$ for all $N \geq 1$ via this refined matching, but the argument requires case analysis at each N.

Corollary 7.5 (The E_n circle partly closes for simple $\mathfrak{g}$). *For simple $\mathfrak{g}$ at non-critical level:*

(i) *The bulk E_3^{top}-algebra (Chern–Simons, arrow origin) maps to the boundary E_2-chiral algebra $V_k(\mathfrak{g})$ (arrow 1).*
(ii) *The ordered bar complex extracts the E_1 data (arrow 2).*
(iii) *The Drinfeld centre recovers E_2 (arrow 3).*

(iv) *The higher Deligne conjecture produces an* E_3^{top}*-algebra* (*arrow* 4).

(v) *The* E_3 *identification* (*Theorem 7.3*) *establishes that this* E_3^{top}*-algebra is isomorphic to the original CFG algebra, closing the circle at the level of formal deformation families.*

Remark 7.6 (Sugawara constraint). The identification (6) carries an additional constraint: the Sugawara element $T_{\mathrm{Sug}} = (1/2(k + h^\vee)) \sum_a :J^a J_a:$ must be Q-exact in the CFG BRST complex. This is verified for $V_k(\mathfrak{g})$ at non-critical level by the explicit antighost contraction (Example 5.2). At the critical level $k = -h^\vee$, the Sugawara element diverges, topologisation fails, and the E_3 identification does not apply.

Remark 7.7 (Extension to $\mathfrak{gl}_N$). For $\mathfrak{g} = \mathfrak{gl}_N$, the E_3 identification extends via two independent invariant bilinear forms B_{tr} and B_{ab}. The deformation base is two-dimensional at each order, parametrised by $(\lambda_1, \lambda_2) \in \mathbb{C}^2$, where $\lambda_1 = k_1 + N$ controls the $\mathfrak{sl}_N$ factor and $\lambda_2 = k_2$ controls the $\mathfrak{gl}_1$ factor. The formal disk comparison determines both scalars independently: choosing $a, b \in \mathfrak{sl}_N$ with $\mathrm{tr}(a) = \mathrm{tr}(b) = 0$ and $\mathrm{tr}(ab) \neq 0$ extracts λ_1; choosing $a = b = I_N/N$ extracts λ_2. The isomorphism

$$\{\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_{k_1,k_2}(\mathfrak{gl}_N))\} \;\simeq\; \{\mathcal{A}^{\lambda_1,\lambda_2}\}$$

holds over the two-dimensional base, order by order in the joint (λ_1, λ_2)-adic filtration.

8. Five notions of E_1-chiral algebra

8.1. **The problem.** The phrase "E_1-chiral algebra" admits at least five inequivalent formalizations. Each captures a different aspect of the same informal idea (an associative algebra object in the chiral world), but they lead to different derived centres, different obstruction theories, and different relationships to the E_n operadic circle. This section catalogues the five notions, states the known relationships, and identifies the open comparisons.

8.2. **The five notions.**

Definition 8.1 (Five notions of E_1-chiral algebra). Let X be a smooth complex curve. An *E_1-chiral algebra on X* can mean any of the following:

(A) *Strict* $\mathrm{Ass}^{\mathrm{ch}}$*-algebra.* An algebra over the associative chiral operad $\mathrm{Ass}^{\mathrm{ch}}$, i.e. a $\mathcal{D}$-module $\mathcal{A}$ on X with a binary product $\mu\colon \mathcal{A} \boxtimes \mathcal{A} \to \Delta_! \mathcal{A}$ satisfying strict associativity: $\mu \circ (\mu \otimes \mathrm{id}) = \mu \circ (\mathrm{id} \otimes \mu)$. This is the $\mathcal{D}$-module-theoretic formulation.

(B) *A_∞-algebra in* $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$. An operad map $A_\infty \to \mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$, where $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$ is the chiral endomorphism operad. The structure maps are $m_k^{\mathrm{ch}}\colon \mathcal{A}^{\otimes k} \to \mathcal{A}((\lambda_1)) \cdots ((\lambda_{k-1}))$ satisfying the Stasheff identities. This is the working definition of the present paper and the monograph. Strictification ($m_k = 0$ for $k \geq 3$) recovers Notion (A).

(C) *Etingof–Kazhdan quantum vertex algebra.* A pair $(\mathcal{A}, S(z))$ where $S(z)$ is a vertex R-matrix satisfying the five EK axioms: S-locality, quantum Yang–Baxter equation, unitarity, shift condition, and hexagon. The R-matrix is the primary datum.

(D) *A_∞-algebra object in E_1-chiral algebras.* A double A_∞ structure: an inner chiral A_∞ from Notion (B) controlling the algebra, and an outer A_∞ controlling higher coherences between E_1-chiral algebra maps. This arises naturally when studying functors and natural transformations in the E_1-chiral world.

(E) *Factorisation algebra on* $\mathrm{Ran}^{\mathrm{ord}}(X)$. The $\mathcal{D}$-module-theoretic incarnation: a factorisation algebra on the ordered Ran space $\mathrm{Ran}^{\mathrm{ord}}(X) = \coprod_{k \geq 1} \mathrm{Conf}_k^{\mathrm{ord}}(X)$, where the ordered factorisation structure encodes the E_1 data geometrically.

Theorem 8.2 (Comparison of notions (B) and (C) on the Koszul locus)**.** *On the Koszul locus* (*the locus where the bar complex has cohomology concentrated in degree* 1)*, Notions* (B) *and* (C) *determine each other via a Drinfeld associator* $\Phi \in \exp(\hat{\mathfrak{t}}_3)$*:*

(i) (B) $\Rightarrow$ (C): *the A_∞ structure maps m_k^{ch} determine the R-matrix $S(z)$ via the degree-2 bar cohomology class, and the Stasheff identities imply the five EK axioms.*

(ii) (C) $\Rightarrow$ (B): *the EK quantum vertex algebra $(\mathcal{A}, S(z))$ determines the A_∞ structure maps by inverting the bar-cobar adjunction: the cobar of the bar coalgebra (with R-twisted differential) recovers the A_∞ structure up to quasi-isomorphism.*

The comparison depends on the choice of associator Φ: different choices give gauge-equivalent A_∞ structures. On BRST cohomology, the A_∞ structure is associator-independent (*the* GRT_1 *action on the* P_3 *bracket is trivial for simple* $\mathfrak{g}$)*. At chain level, the associator dependence is genuine.*

Proof. The direction (B) $\Rightarrow$ (C) is the content of the bar construction: the degree-2 class in $H^*(\bar{B}^{\mathrm{ord}}(\mathcal{A}))$ is the vertex R-matrix $S(z)$, and the Stasheff identity at degree 3 is the quantum Yang–Baxter equation. The remaining EK axioms (unitarity, shift, hexagon) follow from the augmentation, the spectral parameter, and the associativity coherences of the A_∞ structure.

The direction (C) $\Rightarrow$ (B) uses the Etingof–Kazhdan reconstruction theorem: the quantum vertex algebra axioms determine a pro-nilpotent completion of the bar complex, and the cobar of this completion gives the A_∞ structure. The Drinfeld associator Φ enters in the passage from the topological R-matrix (a solution of QYBE on vector spaces) to the chiral R-matrix (a solution on $\mathcal{D}$-modules): the associator controls the regularisation of the formal power series expansion. □

Warning 8.3 (Each notion has its own derived centre)**.** The five notions of E_1-chiral algebra lead to five a priori distinct derived centres:

(A) $\mathcal{Z}^{(A)} = R\mathrm{Hom}_{(\mathrm{Ass}^{\mathrm{ch}})^e}(\mathcal{A}, \mathcal{A})$: the operadic chiral Hochschild cochains of the strict $\mathrm{Ass}^{\mathrm{ch}}$-algebra.

(B) $\mathcal{Z}^{(B)} = C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$: the chiral Hochschild cochain complex, using the A_∞ structure.

(C) $\mathcal{Z}^{(C)} = R\mathrm{Hom}_{(\mathcal{A},S)}(\mathcal{A}, \mathcal{A})$: the derived endomorphisms of $\mathcal{A}$ as an EK quantum vertex algebra.

(D) $\mathcal{Z}^{(D)}$: the homotopy coherent derived centre, using the double A_∞ structure.

(E) $\mathcal{Z}^{(E)} = \int_{\mathrm{Ran}^{\mathrm{ord}}(X)} \mathcal{A}$: the factorisation homology of the ordered factorisation algebra.

The five derived centres are not known to coincide in general. On the Koszul locus, (B) and (C) give the same derived centre (by Theorem 8.2). The comparison of the remaining notions is open.

Remark 8.4 (Symmetric input simplifies everything). For E_∞-chiral algebras (all standard vertex algebras), the five notions collapse: the R-matrix is trivial ($S(z) = \mathrm{id}$), the A_∞ structure is strict ($m_k = 0$ for $k \geq 3$ on cohomology), and the factorisation algebra descends to the unordered Ran space. All five derived centres coincide. The distinction matters only for genuinely E_1-chiral algebras (Yangians, EK quantum vertex algebras).

Remark 8.5 (The monograph uses Notion (B)). Every theorem in this paper stated for "an E_1-chiral algebra $\mathcal{A}$" means a Notion (B) structure unless explicitly qualified. Notion (B) is the natural setting for the bar-cobar adjunction: the A_∞ structure maps are the components of the bar differential, and the Stasheff identities are $d^2 = 0$.

9. The chiral quantum group equivalence

9.1. **The problem.** The ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is a single object from which three structures are recovered: the vertex R-matrix $S(z)$ (braiding data from the degree-2 collision residue), the chiral A_∞-structure maps m_k^{ch} (higher associativity from boundary strata of associahedra), and the chiral coproduct Δ^{ch} (coalgebra data from deconcatenation). The three structures determine each other on the Koszul locus.

9.2. The equivalence theorem.

Theorem 9.1 (Chiral quantum group equivalence)**.** *The following three structures on an E_1-chiral algebra $\mathcal{A}$ on the Koszul locus determine each other, up to the choice of a Drinfeld associator $\Phi \in \exp(\hat{\mathfrak{t}}_3)$:*

(I) R-matrix datum: *E_1-chiral algebra with vertex R-matrix $S(z)$ satisfying S-locality, the quantum Yang–Baxter equation, unitarity, shift condition, and hexagon axiom* (*the Etingof–Kazhdan axioms*)*.*

(II) A_∞ datum: *A_∞-algebra in the chiral endomorphism operad $\mathrm{End}^{\mathrm{ch}}_{\mathcal{A}}$, with structure maps $m_k^{\mathrm{ch}} \colon \mathcal{A}^{\otimes k} \to \mathcal{A}((\lambda_1)) \cdots ((\lambda_{k-1}))$ satisfying the Stasheff identities.*

(III) Coproduct datum: *E_1-chiral algebra with compatible chiral coproduct $\Delta^{\mathrm{ch}} \colon \mathcal{A} \to (\mathcal{A} \hat{\otimes} \mathcal{A})((z))$, coassociative up to conjugation by Φ.*

The equivalences form a triangle:

(i) $\alpha\colon$ (I) $\to$ (II): *The R-matrix determines the A_∞ structure via the ordered bar complex: the degree-k component of the bar differential, twisted by $S(z)$, gives m_k^{ch}.*

(ii) $\beta\colon$ (II) $\to$ (III): *The A_∞ structure determines the coproduct via the deconcatenation coalgebra structure on the bar complex: the coproduct Δ^{ch} is the desuspended deconcatenation Δ of $\bar{B}^{\mathrm{ord}}(\mathcal{A})$, transferred through the bar-cobar quasi-isomorphism.*

(iii) $\gamma\colon$ (III) $\to$ (I): *The coproduct determines the R-matrix via the universal R-matrix of the Drinfeld double: the R-matrix is the image of the canonical element $\sum e_i \otimes e^i \in \mathcal{A} \hat{\otimes} \mathcal{A}^!$ under the coproduct.*

Proof. The functor α is the bar construction: the ordered bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A})$ with R-twisted differential has degree-k components that are the A_∞ structure maps. The Stasheff identities are equivalent to $d^2 = 0$, which holds by the Arnold relation (Theorem 1.13).

The functor β uses the coalgebra structure: the deconcatenation coproduct on $\bar{B}^{\mathrm{ord}}(\mathcal{A}) = T^c(s^{-1}\bar{\mathcal{A}})$ is coassociative (the cofree tensor coalgebra is coassociative). The bar-cobar quasi-isomorphism transfers this coproduct to $\mathcal{A}$, producing Δ^{ch}. The transfer is exact on the Koszul locus (the bar cohomology is concentrated in degree 1, so higher transfer terms vanish).

The functor γ is the universal R-matrix construction: the Drinfeld double $U_{\mathcal{A}} = \mathcal{A} \bowtie \mathcal{A}^!$ carries a canonical element $\mathcal{R} = \sum e_i \otimes e^i$, where $\{e_i\}$ and $\{e^i\}$ are dual bases of $\mathcal{A}$ and $\mathcal{A}^!$ respectively. The image of $\mathcal{R}$ under $\Delta^{\mathrm{ch}} \otimes \mathrm{id}$ gives the vertex R-matrix.

The triangle closes because $\gamma \circ \beta \circ \alpha$ is the identity: starting from $S(z)$, constructing the A_∞ structure, extracting the coproduct, and recovering the R-matrix gives back $S(z)$. □

Remark 9.2 (Associator dependence)*.* The equivalences depend on the choice of Drinfeld associator Φ. Different choices give gauge-equivalent structures: the R-matrices differ by a conjugation, the A_∞ structures differ by an A_∞ quasi-isomorphism, and the coproducts differ by a twist. On BRST cohomology, the bar-side invariants (κ, shadow tower) are associator-free; the cobar-side and coproduct-side carry genuine associator dependence.

9.3. The $\mathfrak{gl}_N$ tower.

Theorem 9.3 ($\mathfrak{gl}_N$ chiral quantum group for all $N \geq 1$)**.** *For every $N \geq 1$, the principal $\mathcal{W}$-algebra $\mathcal{W}_N$ carries a chiral quantum group datum:*

(I) R-matrix: *the Yang R-matrix* $R(u) = u\,I + \Psi\,P$, *where* I *is the* $N \times N$ *identity,* P *is the permutation matrix, and* Ψ *is the coupling parameter.*

(II) Coproduct: *the Drinfeld coproduct* $\Delta_z(T(u)) = T(u) \cdot T(u-z)$ *as* $N \times N$ *matrix multiplication, where* $T(u)$ *is the generating matrix* $T(u) = \sum_{s=0}^{N} \psi_s\, u^{-s}$ *with* $\psi_s = e_s(\Lambda_1, \ldots, \Lambda_N)$ *the elementary symmetric polynomials in the Miura operators.*

(III) RTT relations: *for* $N \geq 2$,

$$R(u-v)\,(T(u) \otimes I)\,(I \otimes T(v)) \;=\; (I \otimes T(v))\,(T(u) \otimes I)\,R(u-v),$$

which is nontrivial precisely when $N \geq 2$ *(at* $N = 1$*, the RTT relation is the scalar identity* $R(u-v) = R(u-v)$*).*

Concrete verifications run through $N = 3$.

Proof. The R-matrix is the Yang solution of the QYBE for $\mathfrak{gl}_N$. The Drinfeld coproduct is an algebra homomorphism of the Yangian $Y(\widehat{\mathfrak{gl}}_N)$: $\Delta_z(T(u)) = T(u) \cdot T(u-z)$ expresses the multiplicativity of the generating matrix under the spectral-parameter shift. The RTT relation is the compatibility of the R-matrix with the coproduct, proved by direct verification at each N: the relation amounts to the commutativity of the monodromy matrices of the KZ connection at separated spectral parameters.

The spectral coassociativity uses shifted parameters: $(\Delta_{z_1} \otimes \mathrm{id}) \circ \Delta_{z_1+z_2} = (\mathrm{id} \otimes \Delta_{z_2}) \circ \Delta_{z_1}$. □

9.4. **Miura universality.**

Theorem 9.4 (Miura cross-term universality). *For all spins* $s \geq 2$*, the spectral coproduct on the spin-*s *primary generator* W_s *of* $\mathcal{W}_{1+\infty}[\Psi]$ *satisfies*

$$\Delta_z(W_s) \;=\; W_s \otimes 1 \;+\; 1 \otimes W_s \;+\; \frac{\Psi - 1}{\Psi}\big(J \otimes W_{s-1} + W_{s-1} \otimes J\big) \;+\; (\text{lower-spin and composite corrections}) \;+\; (\text{spectral tail$$

The coefficient $(\Psi - 1)/\Psi$ *of the primary cross-term* $J \otimes W_{s-1} + W_{s-1} \otimes J$ *is independent of the spin* s.

Proof. The proof uses the Prochazka–Rapcak Miura factorisation $T(u) = \prod_{i=1}^{N}(u + \Lambda_i)$, where the Λ_i are the Miura operators. The Drinfeld coproduct acts on $T(u)$ as matrix multiplication: $\Delta_z(T(u)) = T(u) \cdot T(u-z)$.

The generators $\psi_s = e_s(\Lambda_1, \ldots, \Lambda_N)$ are elementary symmetric polynomials. The Drinfeld formula at u^{-s} gives $\Delta_z(\psi_s) = \sum_{a+b=s} \psi_a \otimes \psi_b$ + (spectral tail). The W-algebra generators $W_s = \psi_s - (1/\Psi) :J\, W_{s-1}: - \cdots$ are obtained by the nonlinear Miura inversion. The coefficient $(\Psi - 1)/\Psi$ on $J \otimes W_{s-1}$ arises as $1 - 1/\Psi$: the Drinfeld coproduct contributes 1 from the cross-term $\psi_1 \otimes \psi_{s-1}$, and the Miura inversion subtracts $1/\Psi$ from the $:J W_{s-1}:$ correction. Since both the cross-term and the Miura correction are s-independent in structure, the coefficient $(\Psi - 1)/\Psi$ is universal. □

Remark 9.5 (Verification and coincidental agreement). The coefficient $(\Psi - 1)/\Psi$ equals $1/\Psi$ at $\Psi = 2$ and $1/2$ at $\Psi = 2$: these coincide. This is a coincidental agreement that masks the distinction between $(\Psi - 1)/\Psi$ and $1/\Psi$ at a single point. Verification requires Ψ-values at three or more points (e.g. $\Psi = 1, 2, 3$). At $\Psi = 1$: $(\Psi - 1)/\Psi = 0$ (the coproduct is an algebra homomorphism, consistent with the commutative limit); $1/\Psi = 1$ (wrong). Concrete verification runs through spins 2–6.

9.5. **Verlinde recovery.**

Proposition 9.6 (Verlinde formula from ordered chiral homology). *At integer level* $k \geq 0$*, the dimension of the ordered chiral homology of* $V_k(\mathfrak{sl}_2)$ *at genus* g *recovers the Verlinde formula:*

$$Z_g \;=\; \sum_{j=0}^{k} S_{0j}^{2-2g}, \qquad S_{0j} = \sqrt{\frac{2}{n}}\,\sin\!\left(\frac{\pi(j+1)}{n}\right), \qquad n = k+2.$$

Closed-form polynomials $P_g(n) = n^{g-1}(n^2-1)\cdot R_{g-2}(n^2)$ are proved through genus 6:

$$Z_0 = 1, \qquad Z_1 = n-1, \qquad Z_2 = \frac{n(n^2-1)}{6},$$
$$Z_3 = \frac{n^2(n^2-1)(n^2+11)}{180}, \qquad Z_4 = \frac{n^3(n^2-1)(2n^4+23n^2+191)}{7560}.$$

The leading coefficients are $\mathrm{leading}(Z_g) \sim \zeta(2g-2)/(2^{g-2}\pi^{2g-2})$*, consistent with the asymptotic growth of modular forms.*

Proof. The Verlinde formula computes the dimension of conformal blocks at genus g and level k. The ordered chiral homology at level k is the factorisation homology of $V_k(\mathfrak{sl}_2)$ over the ordered configuration space on a genus-g surface. Handle attachment (sewing along a pair of pants) and separating factorisation are verified: the pair-of-pants fusion product (Definition 2.13) reproduces the Verlinde fusion coefficients. The closed-form polynomials are obtained by evaluating $\sum_{j=0}^{k} S_{0j}^{2-2g}$ using the explicit S-matrix entries and summing by residue calculus. The rational generating function $G_n(x) = \sum_j 1/(1-a_j x)$ with $a_j = n/(2\sin^2(\pi j/n))$ produces all Z_g as coefficients. □

Remark 9.7 (Conformal anomaly obstruction). The obstruction to a constant (non-spectral) coproduct for the Virasoro algebra is the conformal anomaly: the quartic-pole OPE $T(z)\,T(w) \sim (c/2)/(z-w)^4 + \cdots$ forces a primitive ansatz $\Delta(T) = T\otimes 1 + 1\otimes T + X$ to satisfy $(c/2+c/2)/(z-w)^4$ on the left but $(c/2)/(z-w)^4$ on the right. The mismatch $c/2$ is the conformal anomaly, quantified by $\kappa(\mathrm{Vir}_c) = c/2$. At $c=0$, the obstruction vanishes and a constant coproduct exists. At $c\neq 0$, the spectral parameter is forced.

10. Three computations

10.1. **The programme.** The E_n operadic circle assigns to each chiral algebra $\mathcal{A}$ a package of data at each operadic level: E_1 (bar complex, R-matrix, Yangian), E_2 (derived centre, Gerstenhaber bracket), E_3^{top} (topologised bulk, BV structure), together with the shadow tower (κ, S_4, Δ, class) and the genus anomaly obs_g. The three standard families (Heisenberg, affine Kac–Moody, Virasoro) exhibit the full range of behaviour.

10.2. **Heisenberg H_k: class G.**

Proposition 10.1 (Complete E_n data for H_k). *The Heisenberg algebra H_k at level k has the following E_n data:*

Datum	***Value***	***Source***
OPE	$J(z)J(w)\sim k/(z-w)^2$	*defining*
r-matrix	$r^{\mathrm{Heis}}(z) = k/z$	*Example 1.17*
κ	$\kappa(H_k) = k$	$\mathrm{av}(k/z) = k$
Class	G	$S_4 = 0$, $\Delta = 8k\cdot 0 = 0$
R-matrix	$S(z) = \mathrm{id}$	*abelian*
Yangian	*cocommutative*	*abelian*
Koszul dual	$H_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$, $\kappa^! = -k$	$K = k + (-k) = 0$
ChirHoch*	$\mathbb{C}\oplus\mathbb{C}[-1]\oplus\mathbb{C}[-2]$	*Example 3.7*
E_3^{top}	*trivial (P_3 bracket zero)*	*abelian CS*
SC-formality	*SC-formal*	*class G $\Leftrightarrow$ SC-formal*
Genus anomaly	$\mathrm{obs}_g = k\cdot\lambda_g$ (uniform-weight)	$d_{\mathrm{fib}}^2 = k\,\omega_g$

Proof. The OPE is the defining data. The r-matrix $r(z) = k/z$ is the d log-absorption of the double pole: the OPE $J(z)J(w) \sim k/(z-w)^2$ gives propagator residue $\mathrm{Res}_{z=w} d\log(z-w) \cdot k/(z-w)^2 = k/(z-w)$, hence $r(z) = k/z$. At $k = 0$, $r(z) = 0$: the r-matrix vanishes in the abelian limit. The averaging map gives $\kappa = \mathrm{av}(k/z) = k$. Since $S_4 = 0$ (the algebra has no quartic pole), $\Delta = 8k \cdot 0 = 0$ and the shadow tower terminates at degree 2: class G.

The R-matrix $S(z) = \mathrm{id}$ because the OPE is Σ_k-symmetric (abelian). The Yangian $Y(\widehat{\mathrm{U}(1)})$ is the universal enveloping algebra of the abelian Lie algebra, which is cocommutative.

The chiral Hochschild cohomology has total dimension 3 (Example 3.7). The P_3 bracket on $C^*(H_k)$ is zero because $[J, J] = 0$ (abelian). The E_3^{top} structure is trivial (all operations are determined by the commutative cup product).

SC-formality holds because the shadow tower terminates: class G implies SC-formal (the discriminant $\Delta = 0$ forces all higher SC operations to vanish). □

10.3. **Affine Kac–Moody $V_k(\mathfrak{g})$: class L.**

Proposition 10.2 (Complete E_n data for $V_k(\mathfrak{g})$). *The affine Kac–Moody algebra $V_k(\mathfrak{g})$ at non-critical level $k \neq -h^\vee$, with $\mathfrak{g}$ simple of dimension $d = \dim(\mathfrak{g})$, has the following E_n data:*

Datum	***Value***	***Source***
OPE	$J^a(z)J^b(w) \sim kB^{ab}/(z-w)^2 + f^{ab}{}_c J^c/(z-w)$	*defining*
r-matrix	$r^{\mathrm{KM}}(z) = k\,\Omega/z$ (trace-form)	$d\log$ *absorption*
κ	$\kappa = d(k+h^\vee)/(2h^\vee)$	$\mathrm{av}(r) + d/2$
Class	L	$S_4 = 0$, *shadow depth* $r = 3$
R-matrix	$S(z) = \mathrm{id}$	E_∞*-chiral*
Yangian	$Y(\hat{\mathfrak{g}})$, *cocommutative*	E_∞*-chiral*
Koszul dual	$V_k(\mathfrak{g})^!$, $\varkappa = 0$	$\varkappa = \kappa + \kappa' = 0$
ChirHoch*	$\mathbb{C} \oplus \mathfrak{g}[-1] \oplus \mathbb{C}[-2]$	*Example 3.8*
E_3^{top}	*nontrivial*, $\simeq$ *CFG CS*	*Theorem 7.3*
SC-formality	*not SC-formal*	*class L*, $\ell_3 \neq 0$
Genus anomaly	$\mathrm{obs}_g = \kappa \cdot \lambda_g$ (uniform-weight)	*curvature*

The Koszul complementarity $\kappa + \kappa' = 0$ holds for all simple $\mathfrak{g}$ and all k.

Proof. The r-matrix uses the trace-form convention: $r(z) = k\,\Omega/z$, where Ω is the inverse Killing form Casimir. At $k = 0$, $r(z) = 0$: the r-matrix vanishes (abelian limit). The averaging map gives $\mathrm{av}(r(z)) = k\dim(\mathfrak{g})/(2h^\vee)$ (the double-pole channel contribution); the full κ includes the Sugawara shift $\dim(\mathfrak{g})/2$ from the simple-pole self-contraction: $\kappa = \dim(\mathfrak{g})(k+h^\vee)/(2h^\vee)$. At $k = -h^\vee$ (critical): $\kappa = 0$, the curvature vanishes, and the bar complex is uncurved.

Class L: the OPE has a double pole and a simple pole, giving shadow depth $r = 3$. Since $S_4 = 0$ (the OPE has no quartic pole), $\Delta = 0$ and the shadow tower terminates at degree 3.

The E_3^{top} structure is nontrivial: the P_3 bracket $\{\phi_a, \phi_b\} = (k+h^\vee) f_{ab}{}^c \phi_c$ is the linearisation of the Chern–Simons interaction, and the identification with the CFG algebra is Theorem 7.3.

$\mathrm{ChirHoch}^1(V_k(\mathfrak{g})) \cong \mathfrak{g}$ as a vector space, with total dimension $\dim(\mathfrak{g}) + 2$. □

Remark 10.3 (The Sugawara shift). For the Heisenberg (abelian), $\mathrm{av}(r(z)) = \kappa$ holds directly. For non-abelian Kac–Moody, the full κ includes the Sugawara shift: $\kappa = \mathrm{av}(r(z)) + \dim(\mathfrak{g})/2$. The term $\dim(\mathfrak{g})/2$ arises from the simple-pole self-contraction through the adjoint Casimir eigenvalue $2h^\vee$. This is a fundamental difference between abelian and non-abelian averaging, and is the reason $\kappa(V_k(\mathfrak{g})) \neq 0$ at $k = 0$ (the non-abelian Lie bracket persists).

10.4. **Virasoro Vir_c: class M.**

Proposition 10.4 (Complete E_n data for Vir_c). *The Virasoro algebra* Vir_c *at central charge c has the following E_n data:*

Datum	***Value***	***Source***
OPE	$T(z)T(w) \sim (c/2)/(z-w)^4 + 2T(w)/(z-w)^2 + T'(w)/(z-w)$	*defining*
r-matrix	$r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$	$d\log$ *absorption*
κ	$\kappa(\mathrm{Vir}_c) = c/2$	*census*
Class	M	$S_4 \neq 0$, $d_{\mathrm{alg}} = \infty$
R-matrix	$S(z) = \mathrm{id}$	E_∞*-chiral*
Yangian	$Y^{\mathrm{dg}}(\mathrm{Vir}_c)$, *dg-shifted*	$d_{\mathrm{gen}} = 3$
Koszul dual	Vir_{26-c}, $K = 13$	*self-dual at* $c = 13$
ChirHoch*	*concentrated in* $\{0, 1, 2\}$, *polynomial*	*Theorem 3.5*
E_3^{top}	*proved* (*cohom.* + *model*)	*Theorem 5.10, Proposition 5.13*
SC-formality	*not SC-formal*	*class M*
Genus anomaly	$\mathrm{obs}_g = (c/2)\cdot\lambda_g$ (uniform-weight)	*curvature*

Proof. The r-matrix: the OPE has a quartic pole $(c/2)/(z-w)^4$, a double pole $2T(w)/(z-w)^2$, and a simple pole $T'(w)/(z-w)$. The $d\log$ propagator absorbs one pole order (the r-matrix pole is the OPE pole minus 1): the quartic gives cubic $(c/2)/z^3$, the double gives simple $2T/z$, and the simple pole contributes a regular term. Hence $r(z) = (c/2)/z^3 + 2T/z$, not quartic.

The scalar $\kappa = c/2$ is the unique family for which κ equals half the central charge. The Koszul dual is Vir_{26-c} with $\kappa' = (26-c)/2$, giving the scalar complementarity $\varkappa := \kappa + \kappa' = c/2 + (26-c)/2 = 13$ (distinct from the Trinity ghost-charge conductor $K(\mathrm{Vir}) := c + c^! = 26 = -c_{\mathrm{ghost}}(\mathrm{BRST}_{\mathrm{Vir}})$; related by $\varkappa = \varrho_{\mathrm{Vir}} \cdot K$, $\varrho_{\mathrm{Vir}} = 1/2$). Self-duality at $c = 13$: $\varkappa = 13$ and $\kappa = \kappa' = 13/2$.

Class M: the OPE quartic pole gives $S_4 \neq 0$, hence $\Delta = 8\kappa\, S_4 \neq 0$ and the shadow tower is infinite. The generating depth is $d_{\mathrm{gen}} = 3$ (m_3 generates all higher operations), but the algebraic depth is $d_{\mathrm{alg}} = \infty$ (all m_k are nonzero). The algebra is not SC-formal.

The cohomological E_3^{top} structure for Virasoro is proved by Theorem 5.10: the Costello–Gaiotto theorem provides 3d holomorphic Chern–Simons with DS boundary conditions, and the DS-transported antighost G' satisfies $T_{\mathrm{DS}} = [Q_{\mathrm{CS}}, G']$ on Q_{CS}-cohomology. The chain-level $(E_3)_\infty$ on a quasi-isomorphic model follows from Proposition 5.13. The original-complex lift remains open (class M: infinite coherence tower; Conjecture 5.8(ii)). □

Remark 10.5 (The S_4 invariant). The S_4 invariant of Virasoro is $S_4 = 10/(c(5c+22))$, with large-c asymptotic $S_4 \sim 2/c^2$. The discriminant $\Delta = 8\kappa\, S_4 = 8 \cdot (c/2) \cdot 10/(c(5c+22)) = 40/(5c+22)$ is nonzero for all $c \neq \infty$.

Remark 10.6 (Generating depth vs algebraic depth). The Virasoro illustrates the distinction between generating depth and algebraic depth. The operation m_3 generates all higher m_k by recursive application ($d_{\mathrm{gen}} = 3$: finite), but every m_k is nonzero ($d_{\mathrm{alg}} = \infty$: the tower never terminates). The generating depth is a finiteness statement about the recursion; the algebraic depth is a statement about vanishing. These are different invariants: the Virasoro has finite generating depth but infinite algebraic depth.

Remark 10.7 (Comparison table: G, L, M). The three computations exhibit the escalation from G to L to M:

	Class G (Heis)	Class L (KM)	Class M (Vir)
OPE poles	2	2 (dp) + 1 (sp)	4 + 2 + 1
Shadow depth	$r = 2$	$r = 3$	$r = \infty$
Δ	0	0	$\neq 0$
E_3	trivial	nontrivial (proved)	conjectural
SC-formal	yes	no	no
ChirHoch dim	3	$\dim(\mathfrak{g}) + 2$	polynomial
K	0	0	13

The passage from G to L introduces the non-abelian Lie bracket and the non-trivial P_3 structure. The passage from L to M introduces the quartic pole, the infinite shadow tower, and the gravitational sector.

11. The closing conjecture

11.1. **Statement.** The E_n operadic circle (5) traverses $E_3 \to E_2 \to E_1 \to E_2 \to E_3$. Theorem 7.3 establishes that the output E_3 is isomorphic to the input for simple $\mathfrak{g}$. The closing conjecture asks whether this holds in full generality: not just for the deformation families, but for all E_1-chiral algebras with topologisable bulk.

Conjecture 11.1 (Drinfeld centre equals derived chiral centre)**.** *Let $\mathcal{A}$ be an E_1-chiral algebra* (*Notion B, Definition 8.1*) *and $U_{\mathcal{A}} = \mathcal{A} \bowtie \mathcal{A}^!$ the Drinfeld double. Assume the topologisation hypotheses of Section 5, so that the bulk E_3^{top}-algebra and the derived chiral centre are defined. Then there is an equivalence of E_3^{top}-algebras:*

$$Z(U_{\mathcal{A}}) \;\simeq\; \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}),$$

where Z denotes the Drinfeld centre and $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}$ the derived chiral centre.

11.2. **Three obstructions.** Three obstructions to a proof are identified.

(I) *Pointwise reduction for class M.* For class G and L algebras (Heisenberg, affine Kac–Moody), the Drinfeld centre $Z(U_{\mathcal{A}})$ can be computed stalk-wise: the centre at each point of the Ran space agrees with the chiral Hochschild cochains, and the global assembly is straightforward. For class M algebras (Virasoro, $\mathcal{W}_N$), the A_∞ corrections at degree ≥ 3 are invisible to the stalk-wise centre. The global assembly requires a coherent family of higher homotopies that is not produced by the pointwise computation.

(II) *Strict factorisation for $\mathcal{A}^!$.* The Drinfeld double $U_{\mathcal{A}} = \mathcal{A} \bowtie \mathcal{A}^!$ requires the Koszul dual $\mathcal{A}^!$ to carry a factorisation structure compatible with $\mathcal{A}$. The Koszul dual is defined as an $(\mathsf{SC}^{\mathrm{ch,top}})^!$-algebra (Proposition 4.8): the closed colour is Lie, the open colour is Ass. The passage from the $(\mathsf{SC}^{\mathrm{ch,top}})^!$-algebra to a factorisation algebra on the Ran space requires careful completion in the pro-nilpotent topology. The completion is non-trivial for infinite-dimensional chiral algebras: the pro-nilpotent filtration on $\mathcal{A}^!$ may not be compatible with the factorisation product.

(III) *Bar-cobar vs chiral Hochschild.* Bar-cobar inversion produces the algebra $\mathcal{A}$ from the bar complex $\bar{B}^{\mathrm{ord}}(\mathcal{A})$: $\Omega(\bar{B}^{\mathrm{ord}}(\mathcal{A})) \simeq \mathcal{A}$. This is inversion, not the chiral Hochschild construction. The compatibility between the bar-cobar involution and the chiral Hochschild functor (which computes the derived centre from the bar complex) is an independent statement. Concretely: the convolution $\mathrm{Hom}(\bar{B}^{\mathrm{ord}}(\mathcal{A}), \mathcal{A})$ computes $C^\bullet_{\mathrm{ch}}(\mathcal{A}, \mathcal{A})$, and this must be compared with the Drinfeld centre $Z(\mathcal{A}\text{-mod})$. The comparison requires a Morita-type equivalence between the module category of $\mathcal{A}$ and the comodule category of $\bar{B}^{\mathrm{ord}}(\mathcal{A})$, which is the bar-cobar equivalence of modules.

Theorem 11.2 (Resolution of any one obstruction suffices)**.** *If any one of the three obstructions* (I)–(III) *is resolved, the conjecture reduces to a comparison of two E_2-categories, which is then a consequence of the higher Deligne conjecture.*

Proof. If (I) is resolved: the Drinfeld centre is computed stalk-wise for all classes, so $Z(U_{\mathcal{A}})$ is computed as a chiral Hochschild cochain complex. The comparison with $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is the chiral centre theorem (Theorem 3.3).

If (II) is resolved: the Drinfeld double $U_{\mathcal{A}}$ is a well-defined E_1-chiral algebra, and $Z(U_{\mathcal{A}})$ is a well-defined E_2-braided monoidal category. The derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ is also an E_2-algebra by the chiral centre theorem. The comparison of the two E_2-algebras reduces to matching their P_3-structures via E_3 formality (Theorem 7.1), and the one-dimensionality of the deformation space (for simple $\mathfrak{g}$) forces agreement.

If (III) is resolved: the bar-cobar equivalence of modules produces a Morita equivalence between $\mathcal{A}$-modules and $\bar{B}^{\mathrm{ord}}(\mathcal{A})$-comodules. The Drinfeld centre of the E_1-monoidal category of $\mathcal{A}$-modules is then computed by the chiral Hochschild cochains of $\bar{B}^{\mathrm{ord}}(\mathcal{A})$, which is $C^{\bullet}_{\mathrm{ch}}(\mathcal{A},\mathcal{A}) = \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$ by definition.

In each case, the resulting comparison is between two E_2-algebras with matching P_3-structure. The higher Deligne conjecture promotes both to E_3, and the E_3 identification (Theorem 7.3) establishes the equivalence. □

11.3. **Evidence.**

Proposition 11.3 (Evidence for the closing conjecture)**.** *The following cases are verified:*

(i) Heisenberg H_k, $k \neq 0$ (class G). *The Drinfeld double $U_{H_k} = H_k \bowtie H_k^!$ has centre $Z(U_{H_k}) = \mathbb{C}^3$ (Proposition 10.1), matching the derived centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k) = \mathbb{C}^3$ (Example 3.7). All three obstructions are trivially resolved: pointwise reduction works (class G), $H_k^!$ is a free-field algebra (factorisation automatic), and bar-cobar inversion is a quasi-isomorphism (class G, Koszul).*

(ii) Affine Kac–Moody $V_k(\mathfrak{g})$, simple $\mathfrak{g}$, $k \neq -h^\vee$ (class L). *The E_3 identification theorem (Theorem 7.3) establishes the closing at the level of formal deformation families. The closing conjecture asks for the identification as E_3-algebras, which requires matching the Drinfeld centre with the derived centre. The pointwise reduction obstruction (I) is resolved for class L (the shadow tower terminates at degree 3, so the stalk-wise centre sees all the data). Obstruction (II) requires the Koszul dual $V_k(\mathfrak{g})^!$ to carry factorisation structure; this is proved for $\mathfrak{g} = \mathfrak{sl}_2$.*

(iii) Numerical verification for H_k. *The five invariants of the Drinfeld centre (E_2 dimension, P_3 bracket, cup product, E_3 operation, obstruction class) match at six levels $k = 1, 2, \ldots, 6$. The naive centre has dimension 1; the derived centre has dimension 3 (the* Ext^1 *and* Ext^2 *contributions are invisible to the commutant).*

11.4. **Proof for class G.** The closing conjecture is proved for the Heisenberg algebra by resolving obstruction (III) through the chiral Hochschild–coHochschild duality.

Lemma 11.4 (Chiral Morita–Hochschild comparison for H_k)**.** *Let $C = \bar{B}^{\mathrm{ord}}(H_k)$ and $C^e = C \otimes C^{\mathrm{op}}$. The chiral Drinfeld centre of the E_1-monoidal category of complete H_k-modules admits an identification*

$$Z(H_k\text{-mod}^{\mathrm{compl}}) \;\simeq\; \mathrm{Coext}^*_{C^e}(C_\Delta, C_\Delta) \;\simeq\; \mathrm{ChirHoch}^*(H_k, H_k) \;=\; \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k),$$

where C_Δ is the diagonal bicomodule.

Proof. The argument has three steps.

Step 1: Module–comodule equivalence. The Heisenberg algebra H_k is Koszul with finite-type grading (each weight space is finite-dimensional). By the module Koszul duality theorem, the bar-cobar adjunction restricts to an equivalence of derived categories:

$$D(\mathrm{Mod}_{H_k}^{E_1,\,\mathrm{compl}}) \simeq D(\mathrm{CoMod}_{C}^{E_1,\,\mathrm{conil}}).$$

This equivalence is monoidal with respect to the fusion product on H_k-modules and the cotensor product on C-comodules: the fusion $M_1 \otimes^{\mathbf{L}}_{H_k} M_2$ corresponds to the derived cotensor $\mathcal{K}(M_1) \,\square^{\mathbf{R}}_{C}\, \mathcal{K}(M_2)$. The monoidality follows because the bar resolution is a two-sided coalgebra resolution: the two-pointed bar complex $\bar{B}^{\mathrm{ord}}(H_k;\, M_1, M_2)$ computes the fusion, and under the equivalence this is exactly the cotensor of the corresponding comodules.

Step 2: Drinfeld centre as Hochschild self-extension. The Drinfeld centre of an E_1-monoidal category $\mathcal{C}$ is the category of pairs (M, γ) where γ is a half-braiding. For a module category $\mathcal{C} = A$-mod, the endomorphism algebra of the identity functor is $Z(\mathcal{C}) = \mathrm{Nat}(\mathrm{id}_{\mathcal{C}}, \mathrm{id}_{\mathcal{C}}) = \mathrm{HH}^*(A, A)$. More precisely, the Hochschild cochain complex $C^*(A, A)$ computes the derived endomorphisms of the identity bimodule: $Z(\mathcal{C}) \simeq R\mathrm{Hom}_{A^e}(A, A) = \mathrm{HH}^*(A, A)$. The monoidal equivalence of Step 1 transports this to the coalgebra side: the Drinfeld centre of C-comod is the coHochschild self-extension $\mathrm{Coext}^*_{C^e}(C_\Delta, C_\Delta)$.

Step 3: Hochschild–coHochschild duality. The chiral Hochschild–coHochschild duality (cohomological version) gives

$$\mathrm{ChirHoch}^*(H_k, H_k) \simeq \mathrm{Coext}^*_{C^e}(C_\Delta, C_\Delta).$$

Composing Steps 1–3:

$$Z(H_k\text{-mod}^{\mathrm{compl}}) \simeq Z(C\text{-comod}^{\mathrm{conil}}) \simeq \mathrm{Coext}^*_{C^e}(C_\Delta, C_\Delta) \simeq \mathrm{ChirHoch}^*(H_k, H_k) = \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k).$$

□

Theorem 11.5 (Closing conjecture for class G). *For the Heisenberg algebra H_k at level $k \neq 0$, the E_n operadic circle closes: there is an equivalence of E_3^{top}-algebras*

$$Z(U_{H_k}) \simeq \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k).$$

Both sides are isomorphic to $\mathbb{C}^3$ as graded vector spaces, with trivial P_3 bracket.

Proof. We verify the hypotheses of the closing conjecture and resolve all three obstructions simultaneously.

Obstruction (I): Pointwise reduction. The Heisenberg is class G: the shadow tower terminates at degree 2 ($S_4 = 0$, $\Delta = 0$). The A_∞ corrections at degree ≥ 3 are absent (all higher m_k vanish for $k \geq 3$ because the OPE has only a double pole). The stalk-wise centre at every point of $\mathrm{Ran}(X)$ agrees with the full derived centre. Pointwise reduction holds unconditionally.

Obstruction (II): Factorisation for $H_k^!$. The Koszul dual $H_k^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ is a free-field chiral algebra generated by V^* with the OPE $\alpha^*(z)\alpha^*(w) \sim -k/(z-w)^2$. As a free field, it carries a canonical factorisation structure on $\mathrm{Ran}(X)$: the factorisation product is the symmetric monoidal structure of the symmetric algebra, and the D-module descent to the Ran space is standard for free-field algebras. The Drinfeld double $U_{H_k} = H_k \otimes H_k^!$ is therefore a well-defined chiral algebra with braiding given by the classical r-matrix $r(z) = k/z$.

Obstruction (III): Bar-cobar vs Hochschild. This is the content of Lemma 11.4: the module–comodule equivalence for H_k transports the Drinfeld centre to the coHochschild self-extension of $\bar{B}^{\mathrm{ord}}(H_k)$, which is identified with the chiral Hochschild cohomology by Hochschild–coHochschild duality. The result:

$$Z(H_k\text{-mod}) \simeq \mathrm{ChirHoch}^*(H_k, H_k) = \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k).$$

Comparison of E_3 structures. The derived chiral centre $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(H_k) = \mathbb{C} \oplus \mathbb{C}[-1] \oplus \mathbb{C}[-2]$ carries trivial P_3 bracket (the Gerstenhaber bracket vanishes because H_k is commutative: the Lie bracket $[J, J] = 0$ forces all brackets on Hochschild cohomology to vanish). The Drinfeld centre $Z(U_{H_k})$ also has trivial P_3 bracket (the R-matrix $S(z) = \mathrm{id}$ gives symmetric braiding, hence trivial bracket on the centre). By E_3 formality (Theorem 7.1), the E_3-structure is determined by the P_3-structure on cohomology. Since both P_3-structures are trivial on the same underlying graded vector space $\mathbb{C}^3$, the E_3-algebras are isomorphic.

The topologisation hypotheses are satisfied for H_k at $k \neq 0$: the conformal vector is $T = (1/(2k)){:}JJ{:}$, and the Sugawara construction makes $\mathbb{C}$-translations Q-exact. □

Remark 11.6 (The abelian simplicity). For the Heisenberg, all three obstructions dissolve simultaneously because the algebra is abelian. Obstruction (I) is resolved by the finiteness of the shadow tower (class G). Obstruction (II) is resolved by the free-field character of $H_k^!$, which gives it canonical factorisation. Obstruction (III) is resolved by Koszulness, which makes the module–comodule equivalence exact. The three obstructions are genuinely independent for general algebras: class M algebras (Virasoro) fail obstruction (I), non-Koszul algebras fail obstruction (III), and algebras whose Koszul duals have non-standard factorisation fail obstruction (II). The Heisenberg is the unique intersection where all three vanish.

11.5. **Proof for class L.** The closing conjecture for affine Kac–Moody algebras at non-critical level follows from obstruction (I) resolution combined with the E_3 identification theorem.

Lemma 11.7 (Stalk-wise reduction for class L). *Let $V_k(\mathfrak{g})$ be the universal affine vertex algebra at non-critical level $k \neq -h^\vee$, with $\mathfrak{g}$ simple. The Drinfeld centre $Z(V_k(\mathfrak{g})\text{-mod})$ is computed by the stalk-wise centre at each point of* $\mathrm{Ran}(X)$*: no A_∞ correction at degree ≥ 4 contributes.*

Proof. The shadow tower of $V_k(\mathfrak{g})$ terminates at degree 3 (class L: the OPE has a double pole and a simple pole, giving $S_4 = 0$ and $\Delta = 0$). The A_∞ structure on the bar complex has m_1 (the bar differential), m_2 (the deconcatenation coproduct), and m_3 (the triple collision residue from the simple pole). All $m_k = 0$ for $k \geq 4$. The stalk-wise Drinfeld centre at degree n is computed by the universal enveloping algebra of the truncated A_∞ operations m_1, m_2, m_3. Since no operation at degree ≥ 4 exists, the stalk-wise computation at each point of $\mathrm{Ran}(X)$ sees the complete A_∞ data: no information is lost in the passage from global to stalk-wise. The global assembly is a limit indexed by the Ran space, and each stalk contributes the full centre data. □

Theorem 11.8 (Closing conjecture for class L). *For $V_k(\mathfrak{g})$ with $\mathfrak{g}$ simple and $k \neq -h^\vee$, the E_n operadic circle closes: there is an equivalence of E_3^{top}-algebras*

$$Z(U_{V_k(\mathfrak{g})}) \;\simeq\; \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})).$$

Both sides are isomorphic to the Costello–Francis–Gwilliam E_3^{top}-algebra from perturbative Chern–Simons theory with gauge algebra $\mathfrak{g}$ at coupling $\lambda = k + h^\vee$.

Proof. The proof proceeds via obstruction resolution and two independent comparison paths.

Step 1: Obstruction (I) resolution. By Lemma 11.7, the stalk-wise centre computes the full Drinfeld centre for class L. The chiral centre theorem (Theorem 3.3) then identifies $Z(V_k(\mathfrak{g})\text{-mod})$ with the chiral Hochschild cochains $C^\bullet_{\mathrm{ch}}(V_k(\mathfrak{g}), V_k(\mathfrak{g}))$. By the theorem on resolution of any one obstruction (Theorem 11.2), the closing conjecture reduces to a comparison of E_2-algebras with matching P_3-structures.

Step 2: P_3 comparison. The P_3-structure on $H^*(\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g}))) \cong C^*(\mathfrak{g})$ is determined by the Gerstenhaber bracket $\{\phi_a, \phi_b\} = \lambda\, f_{ab}{}^c\, \phi_c$ (Remark 7.2). The P_3-structure on $Z(U_{V_k(\mathfrak{g})})$ is determined by the braiding of the Drinfeld double. For affine Kac–Moody at non-critical level, the braiding is the KZ braiding, whose classical limit is the r-matrix $r(z) = k\Omega/z$ (trace-form convention). The Drinfeld centre carries the half-braiding $\gamma_N : M \otimes N \to N \otimes M$ induced by the R-matrix $R(z) = 1 + \lambda\,\Omega/z + O(\lambda^2)$, whose first-order P_3 bracket on the centre is $\{\phi_a, \phi_b\} = \lambda\, f_{ab}{}^c\, \phi_c$. The two P_3 brackets agree.

Step 3: E_3 formality closes the circle. By E_3 formality (Theorem 7.1), the E_3-structure is determined by the P_3-structure on cohomology. For simple $\mathfrak{g}$, $\dim H^3(\mathfrak{g}) = 1$, so the formal deformation family over $\lambda H^3(\mathfrak{g})[[\lambda]]$ is one-dimensional at each order (Step 2 of the proof of Theorem 7.3). Since both P_3-brackets are controlled by the same single parameter $\lambda = k + h^\vee$ and agree at each order, the two E_3-algebras are isomorphic as formal deformation families:

$$Z(U_{V_k(\mathfrak{g})}) \;\simeq\; \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(V_k(\mathfrak{g})) \;\simeq\; \mathcal{A}^{\lambda}.$$

The last identification is Theorem 7.3. □

Remark 11.9 (The two paths to the closing for class L). The proof above resolves obstruction (I) and then uses E_3 formality. An alternative path resolves obstruction (III) directly: the bimodule–bicomodule equivalence for $V_k(\mathfrak{g})$ at non-critical level transports the Drinfeld centre to the coalgebraic coHochschild complex, which is identified with $\mathrm{ChirHoch}^*(V_k(\mathfrak{g}))$ by the same Hochschild–coHochschild duality used in the class G proof (Lemma 11.4). The obstruction to this path is that the bar–cobar module equivalence for non-abelian $V_k(\mathfrak{g})$ requires the PBW completion technology of the ordered chiral Koszul duality theory: the conilpotent comodules over $\bar{B}^{\mathrm{ord}}(V_k(\mathfrak{g}))$ must be compared with complete modules over $V_k(\mathfrak{g})$ in the pro-nilpotent topology. On the Koszul locus (which includes all standard affine Kac–Moody algebras at non-critical level), this comparison is exact and the two paths agree.

Remark 11.10 (Obstruction landscape after class G and L). With classes G and L proved, the closing conjecture remains open only for class M (Virasoro, $\mathcal{W}_N$) and for algebras beyond the standard landscape. The surviving obstruction is the conjunction of (I) and the general topologisation conjecture:

(i) For class M, obstruction (I) is genuine: the infinite shadow tower means the stalk-wise centre does not see the A_∞ corrections at degree ≥ 4, and the global assembly requires coherent families of higher homotopies. Resolving this requires understanding the chain-level A_∞-structure of the Drinfeld centre for algebras with infinite shadow tower: a problem equivalent to the chain-level topologisation problem.

(ii) For class M, cohomological topologisation is now proved (Theorem 5.10), and the model-level $(E_3)_\infty$ follows (Proposition 5.13). The remaining gap is the original-complex chain-level lift (Conjecture 5.8(ii)): the infinite A_∞-coherence tower does not terminate, and applications requiring rigid chain-level data on the specific BRST complex remain conditional on the coderived resolution.

The depth gap theorem ($d_{\mathrm{alg}} \in \{0, 1, 2, \infty\}$, gap at 3) implies there is no intermediate class between L and M: the jump from finite shadow tower (classes G, L, C with $d_{\mathrm{alg}} \leq 2$) to infinite tower (class M with $d_{\mathrm{alg}} = \infty$) is discontinuous. The closing conjecture is therefore proved for all Koszul E_∞-chiral algebras with finite shadow tower (classes G, L, C) that satisfy the topologisation hypotheses, and remains open precisely for class M.

11.6. **Resolution of obstruction (III) on the Koszul locus.** The Hochschild–coHochschild strategy used for class G extends to all Koszul chiral algebras, providing a general resolution of obstruction (III).

Proposition 11.11 (Obstruction (III) resolved on the Koszul locus). *Let $\mathcal{A}$ be a Koszul E_∞-chiral algebra with finite-type grading. Set $C = \bar{B}^{\mathrm{ord}}(\mathcal{A})$. Then the chiral Drinfeld centre of complete $\mathcal{A}$-modules is isomorphic to the derived chiral centre as E_2-algebras:*

$$Z(\mathcal{A}\text{-mod}^{\mathrm{compl}}) \;\simeq\; \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A}).$$

Proof. The argument follows the three-step structure of Lemma 11.4, with each step valid on the Koszul locus.

Step 1. The module Koszul duality theorem gives a monoidal equivalence $D(\mathrm{Mod}_{\mathcal{A}}^{\mathrm{compl}}) \simeq D(\mathrm{CoMod}_{C}^{\mathrm{conil}})$ for Koszul algebras with finite-type grading. The monoidality holds because the two-pointed bar complex computes fusion on both sides.

Step 2. The Drinfeld centre of $\mathcal{A}$-mod$^{\mathrm{compl}}$ is transported across the monoidal equivalence to $\mathrm{Coext}^*_{C^e}(C_\Delta, C_\Delta)$: the coalgebraic self-extension of the diagonal bicomodule in the coderived category of C-bicomodules.

Step 3. The chiral Hochschild–coHochschild duality (cohomological version) identifies $\mathrm{Coext}^*_{C^e}(C_\Delta, C_\Delta) \simeq \mathrm{ChirHoch}^*(\mathcal{A}, \mathcal{A}) = \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(\mathcal{A})$.

The E_2-structure is preserved at each step: Step 1 preserves the monoidal structure (hence the centre), Step 2 preserves the derived endomorphisms of the identity (hence the half-braidings), and Step 3 preserves the Gerstenhaber bracket (the brace operations on Hochschild cochains correspond to the composition product on coalgebraic coextensions). □

Remark 11.12 (Scope of the Koszul locus). All standard chiral algebras in the landscape are Koszul: Heisenberg, affine Kac–Moody (all levels), Virasoro, $\mathcal{W}_N$, $\beta\gamma$, bc, lattice vertex algebras. Proposition 11.11 therefore resolves obstruction (III) for the entire standard landscape. The remaining obstructions for class M are (I) the stalk-wise reduction and the general topologisation conjecture: both are problems about the target E_3^{top}-algebra, not about the comparison map.

11.7. The holographic interpretation.

Remark 11.13 (The circle as holographic duality). If the closing conjecture holds, the E_n operadic circle is a mathematical formulation of holographic duality in 3d holomorphic–topological field theory: the bulk E_3^{top}-algebra determines the boundary E_2-chiral algebra, its E_1-bar complex, the Drinfeld centre, and the derived centre, each recovering the next. The passage from the ordered bar to the symmetric bar is the operadic shadow of the passage from the boundary to the bulk. The five theorems of modular Koszul duality (A, B, C, D, H) are the Σ_n-invariant projections of this holographic structure.

Remark 11.14 (The three obstructions as mathematical problems). Each obstruction identifies a concrete mathematical problem:

(I) The class M pointwise reduction requires understanding A_∞ corrections to the Drinfeld centre for algebras with infinite shadow tower. This is related to the chain-level topologisation problem.

(II) The factorisation structure on $\mathcal{A}^!$ requires constructing the Koszul dual as a genuine factorisation algebra, not merely an $(\mathsf{SC}^{\mathrm{ch,top}})^!$-algebra. This is related to the pro-nilpotent completion problem.

(III) The bar-cobar vs chiral Hochschild comparison requires extending the Keller–Lefèvre-Hasegawa theorem to the chiral setting: the derived Morita equivalence between $\mathcal{A}$-modules and $\bar{B}^{\mathrm{ord}}(\mathcal{A})$-comodules must be proved in the chiral world.

Resolution of any one of these would close the E_n circle for all standard families.

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